

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\403-E-001ABCHydrogen Compressor 1st stage interstage Cooler.pvdb

Date of Analysis : Aug 12,2026 9:54am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 2 of 170
 Vessel Design Summary: Step: 30 9:54am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 800.000 mm ID
 Vessel Design Length, Tangent to Tangent 738.65 cm
 Specified Datum Line Distance 5.00 cm
 Shell Side Design Temperature 127 °C
 Channel Side Design Temperature 71 °C
 Shell Side Design Pressure 9.688 kgf/cm²
 Channel Side Design Pressure 5.625 kgf/cm²
 Shop Shell Side Test Pressure 22.105 kgf/cm²
 Field Shell Side Test Pressure 12.594 kgf/cm²
 Shop Channel Side Test Pressure 11.945 kgf/cm²
 Field Channel Side Test Pressure 7.312 kgf/cm²

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 4	...	...	K03017	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 4	...	...	K03017	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl. head	5.705	0.00	30.000	3.2000	Yes	No
Chnl. rear	5.705	0.00	30.000	3.2000	N/A	No
Chnl. rear flange-2	5.625	0.00	19.000	3.2000	N/A	No
Shell flange	9.688	0.00	20.000	0.0000	N/A	No
Shell	9.688	0.00	44.000	3.2000	N/A	No
shell head	9.688	0.00	25.000	3.2000	No	No

Liquid Level: 80.00 cm Dens.: 5.000 kg/m³ Sp. Gr.: 0.005

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 3 of 170
 Vessel Design Summary: Step: 30 9:54am Aug 12,2026

Type	cm	Length cm	Thickness mm	Thickness mm	Internal mm	External mm	Eff	Eff
Ellipse	0.0	5.0	12.0	12.0	5.7	...	1.00	1.00
Cylinder	70.0	70.0	12.0	12.0	5.7	...	1.00	1.00
Body Flg	80.2	9.8	86.0	80.0	75.7	75.7	1.00	1.00
Body Flg	98.7	8.4	68.0	60.0	54.0	54.0	1.00	1.00
Cylinder	728.6	630.0	16.0	16.0	6.0	...	1.00	1.00
Ellipse	733.7	5.0	12.0	10.2	5.9	...	1.00	1.00

Local Stress Analysis Results:

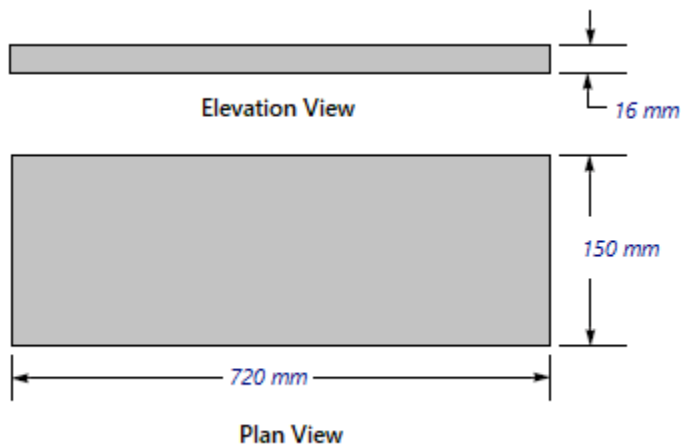
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2- Channel ou	WRC-297	0.673	Shell	Passed
N1- Channel in	WRC-297	0.674	Shell	Passed
Shell inlet	WRC-297	0.987	Shell	Passed
Shell outlet	WRC-297	0.987	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	175.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	660.000 mm
Wear Pad Width	225.000 mm
Wear Pad Thickness	12.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1260.000 mm
Baseplate Length	720.000 mm
Baseplate Thickness	16.000 mm
Baseplate Width	150.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	16.000 mm
Web Thickness	16.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch





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FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 4 of 170
Vessel Design Summary: Step: 30 9:54am Aug 12,2026

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	8606.53	kgf
Maximum Transverse Saddle Shear Load	774.07	kgf
Maximum Longitudinal Saddle Shear Load	785.05	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	8076.51	kgf
Maximum Transverse Saddle Shear Load	255.44	kgf
Maximum Longitudinal Saddle Shear Load	57.50	kgf

Weights:

Fabricated - Bare w/o removable internals	7401.4	kg
Shop Test - Fabricated + water (full)	10864.0	kg
Shipping - Fab. + rem. intls.+ shipping app.	7401.4	kg
Erected - Fab. + rem. intls.+ insul. (etc)	7401.4	kg
Empty - Fab. + intls. + details + wghts.	7455.7	kg
Operating - Empty + operating liquid (no ca)	8568.5	kg
Field Test - Empty weight + test medium (full)	10440.0	kg

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PV Elite 28 Licensee: SL User
 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 5 of 170
 Nozzle Summary: Step: 28 9:54am Aug 12,2026

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2- Channel out	30	...	...	OK	5.70	OK	Passed
N1- Channel inl	30	...	...	OK	5.70	OK	Passed
Shell inlet	27	...	...	OK	5.99	OK	Passed
Shell outlet	27	...	...	OK	5.99	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 27 Nozzle : Shell outlet [Shellside]
 Minimum MAWP Nozzles : 30 Nozzle : N2- Channel out [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N2- Channel outl	350.000	270.000	305.459
20	N1- Channel inle	350.000	90.000	305.459
50	Shell inlet	1536.500	270.000	774.800
50	Shell outlet	6936.500	90.000	774.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 27 9:54am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N2- Channel outl	6.000 in	80	WNF	6.625	10.973	268.27	12.00	270.95	150
N1- Channel inle	6.000 in	80	WNF	6.625	10.973	268.27	12.00	270.95	150
Shell inlet	16.000 in	40	WNF	16.000	12.700	...	...	321.46	150
Shell outlet	16.000 in	40	WNF	16.000	12.700	...	...	321.46	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N2- Channel	SA-106 B	12.000	30.000	10.000	12.000	...
N1- Channel	SA-106 B	12.000	30.000	10.000	12.000	...
Shell inlet	SA-266 2	6.000	10.000	...	...	...
Shell outle	SA-266 2	6.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N2- Channel outl	30.000	270.0	250.00	0.00	Chnl. rear
N1- Channel inle	30.000	90.0	250.00	0.00	Chnl. rear
Shell inlet	148.650	270.0	250.00	0.00	Shell
Shell outlet	688.650	90.0	250.00	0.00	Shell

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PV Elite 28 Licensee: SL User
FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 7 of 170
Bill of Material: Step: 26 9:54am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm THK X 800.0mm ID X 5.0cm	SA-516 70
1	WELD NECK FLANGE: 86.0mm THK X 1020.0mm OD	SA-266 4
1	WELD NECK FLANGE: 68.0mm THK X 1020.0mm OD	SA-266 4
6	BAFFLES: 780.0mm DIA X 6.0mm THK	SA-36
1	TUBESHEET: 1020.0mm X 100.0mm THK	SA-266 4
302	TUBES: 6096.0mm X 25.4mm DIA X 2.5mm THK	SA-179
2	SADDLE: 175.0mm X 120 DEG	SA-516 70
1	INSULATION: 6300mm X 25mm THK	Hot
1	INSULATION: 224mm X 25mm THK	Hot
1	GASKET: 858.0mm OD X 832.0mm ID	...
24	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
2	CYLINDRICAL SEGMENT 12.0mm THK X 998.3mm X 70.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 516.6mm X 70.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 2017.6mm X 630.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 495.7mm X 630.0cm	SA-516 70
1	NAMEPLATE	...



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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 8 of 170
 MDMT Summary: Step: 29 9:54am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell flange [11]	B	-14	-48	-29	0.388	16.000	1.00	No
Shell [8]	B	-14	-48	-29	0.383	16.000	1.00	No
shell head [10]	B	-28	-46	-29	0.690	10.200	1.00	No
shell head [7]	B	-23	-48	-29	0.558	12.000	1.00	No
Left Saddle [24]	B	-23	-48	-29	0.383	12.000	1.00	No
Right saddle [24]	B	-23	-48	-29	0.383	12.000	1.00	No
Shell inlet [1]	B	-5	-104	-29	0.176	11.113	1.00	No
Nozzle Flg [5]	A	-18	-48					
Shell outlet [1]	B	-5	-104	-29	0.176	11.113	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tubesheet: SS[13]	B	-1	-48	-29	0.404	25.000	1.00	No
Warmest MDMT:		-1	-46					
Chnl. rear fl[11]	B	-23	-104	-29	0.303	12.000	1.00	No
Chnl. head [10]	B	-23	-104	-29	0.296	12.000	1.00	No
Chnl. head [7]	B	-23	-104	-29	0.300	12.000	1.00	No
Chnl. rear [8]	B	-23	-104	-29	0.300	12.000	1.00	No
N2- Channel ou[1]	B	-8	-104	-29	0.061	9.601	1.00	No
Nozzle Flg [5]	A	-18	-104					
N1- Channel in[1]	B	-8	-104	-29	0.062	9.601	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: CS[14]	B	-1	-48	-29	0.404	25.000	1.00	No
Bolting [21]		-48						
Warmest MDMT:		-1	-48					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-46.0	11.4	Pass
Channel/Tube	-48.0	11.4	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



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FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 9 of 170
MDMT Summary: Step: 29 9:54am Aug 12,2026

UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:
Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 10 of 170
Warnings and Errors: Step: 0 9:54am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: Chnl. rear flange and Shell flange

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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PV Elite 28 Licensee: SL User
 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 11 of 170
 Input Echo: Step: 1 9:54am Aug 12,2026

Units used in this Analysis (Kgfcm2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	9.6876	kgf/cm ²
Channel Side Design Pressure	5.6249	kgf/cm ²
Shell Side Design Temperature	127.0	°C
Channel Side Design Temperature	71.1	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Severe Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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 Input Echo: Step: 1 9:54am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	



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Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



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Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Chnl. head
Distance "FROM" to "TO"	5 cm
Inside Diameter	800 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5.6249 kgf/cm ²
Design Temperature Internal Pressure	71 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2495.9 kgf/cm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Cooling water
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	80 cm
Liquid Density	999.54 kg/m ³

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Chnl. rear
Distance "FROM" to "TO"	70 cm
Inside Diameter	800 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5.6249 kgf/cm ²
Design Temperature Internal Pressure	71 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid



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Detail ID	Cooling water	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	80	cm
Liquid Density	999.54	kg/m³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2- Channel out1	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	26.293	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1- Channel inle	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	26.293	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Weight	
Detail ID	Partition plate	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	69.999	kgf
Offset from Element Centerline	0	mm

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	Chnl. rear flange-2	
Distance "FROM" to "TO"	9.8	cm
Flange Inside Diameter	800	mm
Element Thickness	80	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	86	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5.6249	kgf/cm²
Design Temperature Internal Pressure	71	°C
Design External Pressure	0	kgf/cm²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 4	
Allowable Stress, Ambient	1407.1	kgf/cm²
Allowable Stress, Operating	1407.1	kgf/cm²
Allowable Stress, Hydrotest	2294.2	kgf/cm²
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	2365.1	kgf/cm²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	



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UNS Number	K03017
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Flange
Description	Shell flange
Distance "FROM" to "TO"	8.4 cm
Flange Inside Diameter	800 mm
Element Thickness	60 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.6875 kgf/cm ²
Design Temperature Internal Pressure	127 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 4
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	630 cm
Inside Diameter	800 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.6875 kgf/cm ²
Design Temperature Internal Pressure	127 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2393.8 kgf/cm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	126 cm
Width of Saddle	175 mm



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Height of Saddle at Bottom	660	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	225	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	504	cm
Width of Saddle	175	mm
Height of Saddle at Bottom	660	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	225	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Hydrogen	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	80	cm
Liquid Density	5	kg/m³

Element From Node	50	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	630	cm
Thickness of Insulation	25	mm
Density	0.00012	kg/cm³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell inlet	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	16	in.
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	182.03	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell outlet	
Dist. from "FROM" Node / Offset dist	590	cm
Nozzle Diameter	16	in.
Nozzle Schedule	40	



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Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	182.03	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	shell head	
Distance "FROM" to "TO"	5	cm
Inside Diameter	800	mm
Element Thickness	10.2	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.6875	kgf/cm ²
Design Temperature Internal Pressure	127	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Hydrogen	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	80	cm
Liquid Density	5	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	22.5	cm
Thickness of Insulation	25	mm
Density	0.00012	kg/cm ³



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XY Coordinate Calculations: Step: 2 9:54am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Chnl. head		5	...	5	...
Chnl. rear		75	...	70	...
Chnl. rear flan		85.25	...	9.8	...
Shell flange		103.65	...	8.4	...
Shell		733.65	...	630	...
shell head		738.65	...	5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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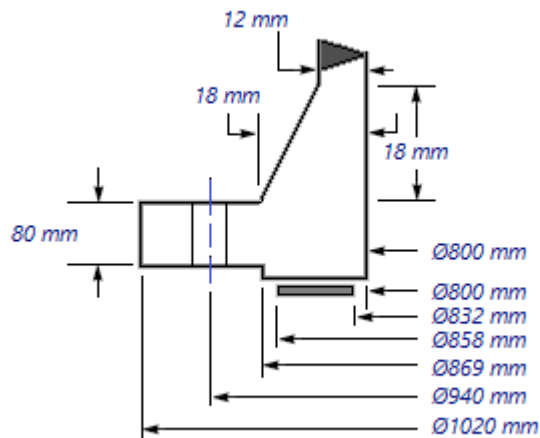
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Flg Calc [Int P]: New Flange Flng: 11 9:54am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl. rear flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	5.62	kgf/cm ²
Design Temperature		71	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	800.000	mm
Flange Outside Diameter	A	1020.000	mm
Flange Thickness	t	80.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	18.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	940.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	UNC Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	869.000	mm
Flange Face Inside Diameter	Fid	800.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	858.000	mm
Gasket Inside Diameter	Gi	832.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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 Flg Calc [Int P]: New Flange Flng: 11 9:54am Aug 12,2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5.62 \cdot (800.0/2 + 3.2)) / (1407.13 \cdot 1.0 - 0.6 \cdot 5.62) + Ca$$

$$= 4.8156 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.8) / (403.2 + 0.6 \cdot 8.8)$$

$$= 30.314 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.0) / (400.0 + 0.6 \cdot 12.0)$$

$$= 41.468 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	76.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	806.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	14.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	8.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	845.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 845.1509^2 \cdot 5.625$$

$$= 31555.252 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 845.1509 \cdot 3.0 \cdot 5.62$$

$$= 5756.953 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 806.4^2 \cdot 5.6249$$

$$= 28727.916 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 31555 - 28728$$

$$= 2827.335 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(31555 + 5757 + 0, 0)$$

$$= 37312.207 \text{ kgf}$$

$$= 64261.406 \text{ kgf, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703.07 \cdot 6.4246 \cdot \pi \cdot 845.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 119929.539 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((64261 + 0 + 4 \cdot 0/845.15)/1754, 119930/1754)$$

$$= 68.382 \text{ cm}^2$$



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 Flg Calc [Int P]: New Flange Flng: 11 9:54am Aug 12,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 76.8/(3.0 + 0.5)$$

$$= 182.457 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 940.0 \cdot \sin(\pi/24)$$

$$= 122.695 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{122.695/(2 \cdot 25.4 + 76.8)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	68.382	85.316	
Radial Distance between Hub and Bolts:	34.925	52.000	
Radial Distance between Bolts and Edge:	26.987	40.000	
Circ. Spacing between the Bolts:	57.150	122.695	182.457

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 85.316 \cdot 1753.82 / (703.07 \cdot \pi(858.0 + 832.0))$$

$$= 4.008 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1753.82(68.382 + 85.316) / 2$$

$$= 134779.06 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 64261 - 31555$$

$$= 32706.16 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (940.0 - 845.1509) / 2$$

$$= 47.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((940.0 - 806.4) / 2 + 47.4245)$$

$$= 57.1123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (940.0 - 806.4 - 14.8) / 2$$

$$= 59.4000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
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 Flg Calc [Int P]: New Flange Flng: 11 9:54am Aug 12,2026

End Pressure, Md	28728.	59.4000	1.0000	1706.
Face Pressure, Mt	2827.	57.1123	1.0000	161.
Gasket Load, Mg	32706.	47.4245	1.0000	1551.
Gasket Seating, Matm	134779.	47.4245	1.0000	6392.

Total Moment for Operation, Mo 3419. kgf-m.
 Total Moment for Gasket seating, Mg 6392. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 84.240 mm
 Hub Ratio, h/h0 = HL / H0 0.214
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.682

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895
 Factor V per 4.16.5 0.382
 Factor f per 4.16.5 1.820

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1020.0/806.4
 = 1.265

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
 = (0.66845 + (5.7169(1.265² • log10(1.265))) / (1.265² - 1))/(1.265 - 1)
 = 8.397

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.265²(1 + 8.55246 • log10(1.265)) - 1) / ((1.0472 + 1.9448 • 1.265²) • (1.265 - 1))
 = 1.812

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
 = (1.265²(1 + 8.55246 • log10(1.265)) - 1) / (1.36136(1.265² - 1) • (1.265 - 1))
 = 9.228

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
 = (1.265² + 1) / (1.265² - 1)
 = 4.334

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 9.228 • 8.8² • 84.24/0.382
 = 157492.953 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.895/84.24
 = 0.011 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (76.8 • 0.011 + 1) / 1.812 + 76.8³/157492.953
 = 3.878

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.8201•3419/806.4)/(3.8783•14.8²)
 = 908.41 kgf/cm²



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 Flg Calc [Int P]: New Flange Flng: 11 9:54am Aug 12,2026

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$
 $= (1.8201 \cdot 6392 / 806.4) / (3.8783 \cdot 14.8^2)$
 $= 1698.29 \text{ kgf/cm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$
 $= (2.0881 \cdot 3419 / 806.4) / (3.8783 \cdot 76.8^2)$
 $= 38.70 \text{ kgf/cm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$
 $= (2.0881 \cdot 6392 / 806.4) / (3.8783 \cdot 76.8^2)$
 $= 72.35 \text{ kgf/cm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$
 $= (8.3975 \cdot 3419 / (76.8^2 \cdot 806.4)) - 4.3338 \cdot 39$
 $= 435.91 \text{ kgf/cm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$
 $= (8.3975 \cdot 6392 / (76.8^2 \cdot 806.4)) - 4.3338 \cdot 72$
 $= 814.94 \text{ kgf/cm}^2$

Average Flange Stress, Operating [SAo]:
 $= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$
 $= (908 + \max(39, 436)) / 2$
 $= 672.16 \text{ kgf/cm}^2$

Average Flange Stress, Seating [SAa]:
 $= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$
 $= (1698 + \max(72, 815)) / 2$
 $= 1256.61 \text{ kgf/cm}^2$

Bolt Stress, Operating [BSo]:
 $= \text{Wm1} / \text{Ab}$
 $= 64261 / 85.316$
 $= 753.22 \text{ kgf/cm}^2$

Bolt Stress, Seating [BSa]:
 $= (\text{Wm2} / \text{Ab})$
 $= (119930 / 85.316)$
 $= 1405.71 \text{ kgf/cm}^2$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	908.41	2110.70	1698.29	2110.70
Radial Flange	38.70	1407.13	72.35	1407.13
Tangential Flange	435.91	1407.13	814.94	1407.13
Maximum Average	672.16	1407.13	1256.61	1407.13
Bolting	753.22	1753.82	1405.71	1753.82

Minimum Required Flange Thickness 75.743 mm

Estimated M.A.W.P. (Operating) 19 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 200.3 kg
 Estimated Unfinished Weight of Forging at given Thk 238.9 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:



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$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6391.8 / 1.0 \cdot 100000.023 \cdot 0.382 / (3.398 \cdot 2067026 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$$

$$= 0.926 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 3419.0 / 1.0 \cdot 100000.023 \cdot 0.382 / (3.398 \cdot 2037826 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$$

$$= 0.502 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6391.8 / 1.0 \cdot 100000.023 \cdot 0.382 / (3.878 \cdot 2067026 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$$

$$= 0.812 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 3419.0 / 1.0 \cdot 100000.023 \cdot 0.382 / (3.878 \cdot 2037826 \cdot 8.8^2 \cdot 84.24 \cdot 0.3)$$

$$= 0.440 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.303, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	780.53	2110.70	1438.41	2110.70
Radial Flange	50.27	1407.13	92.63	1407.13
Tangential Flange	339.81	1407.13	626.21	1407.13
Maximum Average	560.17	1407.13	1032.31	1407.13
Bolting	753.22	1753.82	1405.71	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 66.954 mm

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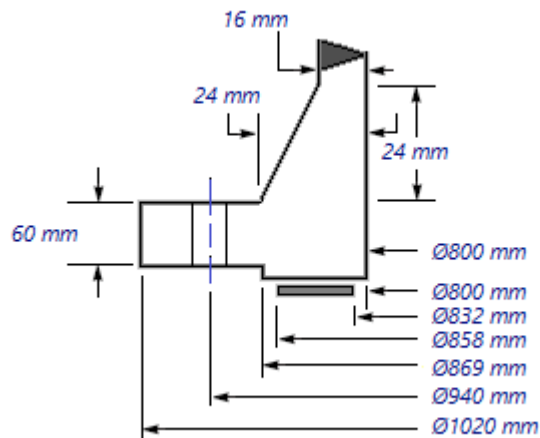
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 Flg Calc [Int P]: New Flange Flng: 12 9:54am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Shell flange

Flange Type	Integral Weld Neck		
Design Pressure	P	9.69	kgf/cm ²
Design Temperature		127	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	800.000	mm
Flange Outside Diameter	A	1020.000	mm
Flange Thickness	t	60.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	g1	24.0000	mm
Length of Hub	h	24.0000	mm
Flange Material	SA-266 4		
Flange Material UNS number	K03017		
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	940.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	UNC Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	869.000	mm
Flange Face Inside Diameter	Fid	800.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	858.000	mm
Gasket Inside Diameter	Gi	832.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.69 \cdot (800.0/2 + 0.0)) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.69) + Ca$$

$$= 2.7653 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 16.0) / (400.0 + 0.6 \cdot 16.0)$$

$$= 54.966 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 16.0) / (400.0 + 0.6 \cdot 16.0)$$

$$= 54.966 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	800.000 mm
Corroded Large Hub,	g1Cor = g1-ci	24.000 mm
Corroded Small Hub,	g0Cor = go-ci	16.000 mm
Code R Dimension,	R = (C-B)/2 - g1	46.000 mm
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm
Basic Gasket Width,	bo = N/2	6.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter,	G = Go - 2 * b	845.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 845.1509^2 \cdot 9.688$$

$$= 54346.422 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 845.1509 \cdot 3.0 \cdot 9.69$$

$$= 9914.984 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 800.0^2 \cdot 9.6875$$

$$= 48694.777 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 54346 - 48695$$

$$= 5651.643 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(54346 + 9915 + 0, 0)$$

$$= 64261.406 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703.07 \cdot 6.4246 \cdot \pi \cdot 845.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 119929.539 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((64261 + 0 + 4 \cdot 0 / 845.15) / 1754, 119930 / 1754)$$

$$= 68.382 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$



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$$= 2 \cdot 25.4 + 6 \cdot 60.0 / (3.0 + 0.5)$$

$$= 153.657 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 940.0 \cdot \sin(\pi / 24)$$

$$= 122.695 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{(Bs / (2a + t))}, 1)$$

$$= \max(\sqrt{(122.695 / (2 \cdot 25.4 + 60.0))}, 1)$$

$$= 1.0523$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	68.382	85.316	
Radial Distance between Hub and Bolts:	34.925	46.000	
Radial Distance between Bolts and Edge:	26.987	40.000	
Circ. Spacing between the Bolts:	57.150	122.695	153.657

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 85.316 \cdot 1753.82 / (703.07 \cdot \pi(858.0 + 832.0))$$

$$= 4.008 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1753.82(68.382 + 85.316) / 2$$

$$= 134779.06 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 64261 - 54346$$

$$= 9914.99 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (940.0 - 845.1509) / 2$$

$$= 47.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((940.0 - 800.0) / 2 + 47.4245)$$

$$= 58.7123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (940.0 - 800.0 - 24.0) / 2$$

$$= 58.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	48695.	58.0000	1.0523	2972.
Face Pressure, Mt	5652.	58.7123	1.0523	349.



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Gasket Load, Mg	9915.	47.4245	1.0523	495.
Gasket Seating, Matm	134779.	47.4245	1.0523	6726.

Total Moment for Operation, Mo 3816. kgf-m.
 Total Moment for Gasket seating, Mg 6726. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 113.137 mm
 Hub Ratio, h/h0 = HL / H0 0.212
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.500

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.898
 Factor V per 4.16.5 0.407
 Factor f per 4.16.5 1.459

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1020.0/800.0
 = 1.275

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
 = (0.66845 + (5.7169(1.275² • log10(1.275))) / (1.275² - 1))/(1.275 - 1)
 = 8.130

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.275²(1 + 8.55246 • log10(1.275)) - 1) / ((1.0472 + 1.9448 • 1.275²) • (1.275 - 1))
 = 1.808

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
 = (1.275²(1 + 8.55246 • log10(1.275)) - 1) / (1.36136(1.275² - 1) • (1.275 - 1))
 = 8.934

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
 = (1.275² + 1) / (1.275² - 1)
 = 4.197

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 8.934 • 16.0² • 113.137/0.407
 = 635888.500 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.898/113.137
 = 0.008 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (60.0 • 0.008 + 1) / 1.808 + 60.0³/635889
 = 1.156

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.4593•3816/800.0)/(1.1562•24.0²)
 = 1045.23 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
 = (1.4593•6726/800.0)/(1.1562•24.0²)



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 Flg Calc [Int P]: New Flange Flng: 12 9:54am Aug 12,2026

= 1842.35 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= ($\beta \cdot Mop / Bcor$) / ($L \cdot t^2$)
 = (1.635•3816/800.0)/(1.1562•60.0²)
 = 187.37 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= ($\beta \cdot Matm/Bcor$) / ($L \cdot t^2$)
 = (1.635•6726/800.0)/(1.1562•60.0²)
 = 330.26 kgf/cm²

Tangential Flange Stress, Operating [STo]:

= ($Y \cdot Mo / (t^2 \cdot Bcor)$) - $Z \cdot SRo$
 = (8.1301•3816/(60.0²•800.0))-4.1968•187
 = 290.89 kgf/cm²

Tangential Flange Stress, Seating [STa]:

= ($y \cdot Matm / (t^2 \cdot Bcor)$) - $Z \cdot SRa$
 = (8.1301•6726/(60.0²•800.0))-4.1968•330
 = 512.73 kgf/cm²

Average Flange Stress, Operating [SAo]:

= ($SHo + \max(SRo, STo)$) / 2
 = (1045+max(187,291))/2
 = 668.06 kgf/cm²

Average Flange Stress, Seating [SAa]:

= ($SHa + \max(SRa, STa)$) / 2
 = (1842+max(330,513))/2
 = 1177.54 kgf/cm²

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$
 = 64261/85.316
 = 753.22 kgf/cm²

Bolt Stress, Seating [BSa]:

= ($Wm2 / Ab$)
 = (119930/85.316)
 = 1405.71 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1045.23	2110.70	1842.35	2110.70
Radial Flange	187.37	1407.13	330.26	1407.13
Tangential Flange	290.89	1407.13	512.73	1407.13
Maximum Average	668.06	1407.13	1177.54	1407.13
Bolting	753.22	1753.82	1405.71	1753.82

Minimum Required Flange Thickness 54.026 mm

Estimated M.A.W.P. (Operating) 20 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 155.8 kg

Estimated Unfinished Weight of Forging at given Thk 204.7 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = 52.14•6915.2/1.0819•100000.023•0.407/(1.038•2067026•16.0²•113.137•0.3)
 = 0.727 (should be ≤ 1)



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 Flg Calc [Int P]: New Flange Flng: 12 9:54am Aug 12,2026

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 3923.3 / 1.0819 \cdot 100000.023 \cdot 0.407 / (1.038 \cdot 2003546 \cdot 16.0^2 \cdot 113.137 \cdot 0.3)$
 $= 0.425$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 6726.2 / 1.0523 \cdot 100000.023 \cdot 0.407 / (1.156 \cdot 2067026 \cdot 16.0^2 \cdot 113.137 \cdot 0.3)$
 $= 0.653$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 3816.0 / 1.0523 \cdot 100000.023 \cdot 0.407 / (1.156 \cdot 2003546 \cdot 16.0^2 \cdot 113.137 \cdot 0.3)$
 $= 0.382$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.388, Temperature Reduction per Fig. UCS 66.1 = 54 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1045.23	2110.70	1842.35	2110.70
Radial Flange	187.37	1407.13	330.26	1407.13
Tangential Flange	290.89	1407.13	512.73	1407.13
Maximum Average	668.06	1407.13	1177.54	1407.13
Bolting	753.22	1753.82	1405.71	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 54.026 mm

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 Internal Pressure Calculations: Step: 5 9:54am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Chnl. head		5.7052	12	3.2	800	1407.1	1
Chnl. rear		5.7052	12	3.2	800	1407.1	2
Chnl. rear flang		5.6249	86	3.2	800	1407.1	3
Shell flange		9.6875	68	...	800	1407.1	4
Shell		9.6879	16	3.2	800	1407.1	5
shell head		9.6879	12	3.2	800	1407.1	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Chnl. head		5.62489	30.234	42.0878	12	5.7	1
Chnl. rear		5.62489	30.234	41.4676	12	5.7	2
Chnl. rear flang		5.62489	18.8048	22.374	80	75.7428	3
Shell flange		9.68753	19.5626	19.5626	60	54.0258	4
Shell		9.68753	43.8356	54.9662	16	5.98749	5
shell head		9.68753	24.6439	35.7907	10.2	5.94884	6

Warning: Shell Side MAWP governed by a standard Flange!

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 17 kgf/cm²
 Shell Side MAPnc = 20 kgf/cm²
 Channel Side MAWP = 9 kgf/cm²
 Channel Side MAPnc = 20 kgf/cm²

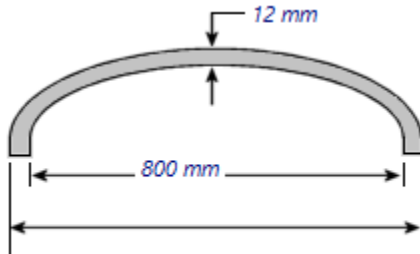
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Chnl. head

Elliptical Head Design Information:			
Design Pressure P	5.705 kgf/cm ²	Int. Design Temperature	71 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	5.000 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	1407.1 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm ²
Inside Diameter Di	800.000 mm	Actual Stress	259.2 kgf/cm ²
Spec. Min. Thk t,e, or sa	12.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	8654.3 cm ²	SG of Contents	1.000
Element Volume	92153.6 cm ³	Weight of Contents	92.1 kgf
Empty Weight	85.3 kgf	Operating Weight	177.4 kgf
		Comp. All. Ope. Stress	1343.1 kgf/cm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5.705 \cdot 806.4 \cdot 0.99) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 5.705)$$

$$= 1.6183 + 3.2000 = 4.8183 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.080 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 8.8) / (0.99 \cdot 806.4 + 0.2 \cdot 8.8)$$

$$= 31 - 0.1 = 31 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$



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$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 800.0 + 0.2 \cdot 12.0)$$

$$= 42 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5.705 \cdot (0.99 \cdot 806.4 + 0.2 \cdot 8.8)) / (2 \cdot 1.0 \cdot 8.8)$$

$$= 259.238 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5.705 \cdot 403.2) / (1407.13 \cdot 1.0 - 0.6 \cdot 5.705) + 3.2$$

$$= 4.839 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.080 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.8) / (403.2 + 0.6 \cdot 8.8)$$

$$= 30 - 0.1 = 30 \text{ kgf/cm}^2$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (806.4 / (2 \cdot 203.2))^2) / 6$$

$$= 0.989543$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.338 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.0, tr = 2.607, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.296$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12.0, tr = 2.643, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.188 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.300	Temperature Difference	81 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

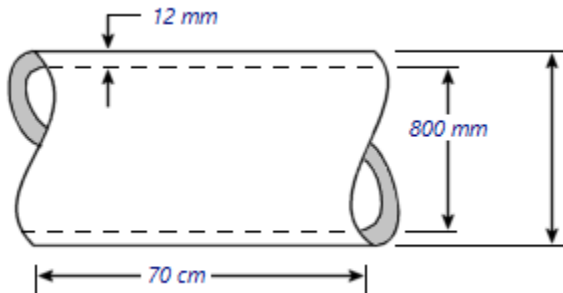
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm



Internal Pressure Calculation Results:

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Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:					
Design Pressure	P	5.705 kgf/cm²	Int. Design Temperature	71 °C	
Material	SA-516 70		Ext. Design Temperature	40 °C	
Length	70.000 cm		External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.2000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	800.000 mm	Actual Stress	264.8 kgf/cm²	
Spec. Min. Thk t,e, or sa	12.0000 mm		Spec. Nominal Thk.	12.000 mm	
Surface Area	18120.7 cm²		SG of Contents	1.000	
Element Volume	351859.2 cm³		Weight of Contents	351.7 kgf	
Empty Weight	166.1 kgf		Operating Weight	517.8 kgf	
			Comp. All. Ope. Stress	1343.1 kgf/cm²	
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency	100 %		Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5.705 \cdot 403.2) / (1407.13 \cdot 1.0 - 0.6 \cdot 5.705) \\
 &= 1.6387 + 3.2000 = 4.8387 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.080 kgf/cm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1407.13 \cdot 1.0 \cdot 8.8) / (403.2 + 0.6 \cdot 8.8) \\
 &= 30 - 0.1 = 30 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1407.13 \cdot 1.0 \cdot 12.0) / (400.0 + 0.6 \cdot 12.0) \\
 &= 41 \text{ kgf/cm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$



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$$= (5.705 \cdot (403.2 + 0.6 \cdot 8.8)) / (1.0 \cdot 8.8)$$
$$= 264.824 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.478 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 12.0$, $t_r = 2.643$, $c = 3.2$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

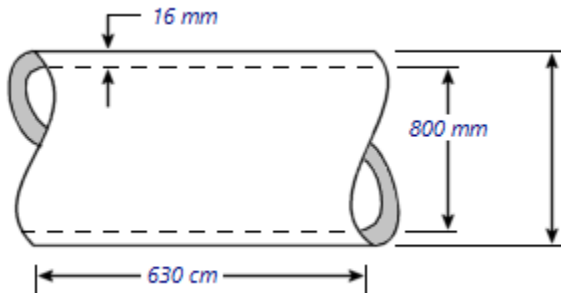
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.188 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.300	Temperature Difference	81 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	9.688 kgf/cm ²	Int. Design Temperature	127 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	630.000 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	1407.1 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm ²
Inside Diameter Di	800.000 mm	Actual Stress	311.0 kgf/cm ²
Spec. Min. Thk t _e , or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	164669.7 cm ²	SG of Contents	0.005
Element Volume	2170857.5 cm ³	Weight of Contents	10.9 kgf
Empty Weight	2002.7 kgf	Operating Weight	2013.6 kgf
		Comp. All. Ope. Stress	1418.6 kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (9.688 \cdot 403.2) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.688) \\
 &= 2.7875 + 3.2000 = 5.9875 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 kgf/cm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1407.13 \cdot 1.0 \cdot 12.8) / (403.2 + 0.6 \cdot 12.8) \\
 &= 44 - 0 = 44 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1407.13 \cdot 1.0 \cdot 16.0) / (400.0 + 0.6 \cdot 16.0) \\
 &= 55 \text{ kgf/cm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (9.688 \cdot (403.2 + 0.6 \cdot 12.8)) / (1.0 \cdot 12.8) \\
 &= 310.983 \text{ kgf/cm}^2
 \end{aligned}$$



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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.961 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 16.0$, $t_r = 4.908$, $c = 3.2$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.383$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

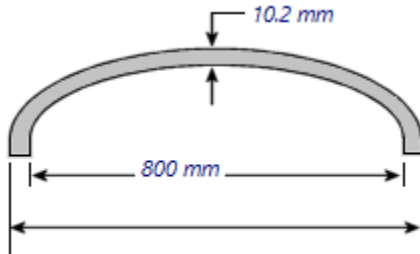
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	17.004 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.383	Temperature Difference	34 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell head

Elliptical Head Design Information:			
Design Pressure P	9.688 kgf/cm ²	Int. Design Temperature	127 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	5.000 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	1407.1 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm ²
Inside Diameter Di	800.000 mm	Actual Stress	553.2 kgf/cm ²
Spec. Min. Thk t,e, or sa	10.2000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	8654.3 cm ²	SG of Contents	0.005
Element Volume	92153.6 cm ³	Weight of Contents	0.5 kgf
Empty Weight	85.3 kgf	Operating Weight	85.7 kgf
		Comp. All. Ope. Stress	1418.6 kgf/cm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (9.688 \cdot 806.4 \cdot 0.99) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 9.688)$$

$$= 2.7488 + 3.2000 = 5.9488 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 7.0) / (0.99 \cdot 806.4 + 0.2 \cdot 7.0)$$

$$= 25 - 0 = 25 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 10.2) / (1.0 \cdot 800.0 + 0.2 \cdot 10.2)$$

$$= 36 \text{ kgf/cm}^2$$



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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (9.688 \cdot (0.99 \cdot 806.4 + 0.2 \cdot 7.0)) / (2 \cdot 1.0 \cdot 7.0)$$

$$= 553.158 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (9.688 \cdot 403.2) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.688) + 3.2$$

$$= 5.987 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.000 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.8) / (403.2 + 0.6 \cdot 8.8)$$

$$= 30 - 0 = 30 \text{ kgf/cm}^2$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (806.4 / (2 \cdot 203.2)))^2 / 6$$

$$= 0.989543$$

Percent Elong. per UCS-79, VIII-1-01-57 (75*tnom/Rf)*(1-Rf/Ro) 6.338 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 10.2, tr = 4.827, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.69, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12.0, tr = 4.908, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.558, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	17.004 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.200 mm	Basic MDMT	-28 °C
Stress/Thk. Ratio	0.690	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-46 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.000 mm



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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	22.105 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	12.594 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.351 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	18.704 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	13.853 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	13.853 kgf/cm ²
App 27-4	= M.A.W.P.	17.004 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.945 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.312 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.983 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	10.107 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	8.044 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8.044 kgf/cm ²
App 27-4	= M.A.W.P.	9.188 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Chnl. head	546.4	2386.0	0.229	12.02
Chnl. rear	558.2	2386.0	0.234	12.02
Shell	712.1	2386.0	0.298	22.18
shell head	1266.7	2386.0	0.531	22.18

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2- Channel outl	Nozzle	1203.20	1203.20	1.000
N2- Channel outl	Pad	1407.13	1407.13	1.000
N1- Channel inle	Nozzle	1203.20	1203.20	1.000
N1- Channel inle	Pad	1407.13	1407.13	1.000
Shell inlet	Nozzle	1407.13	1407.13	1.000
Shell outlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Chnl. head	1407.13	1407.13	1.000
Chnl. rear	1407.13	1407.13	1.000
Chnl. rear flange-2	1407.13	1407.13	1.000
Shell flange	1407.13	1407.13	1.000
Shell	1407.13	1407.13	1.000
shell head	1407.13	1407.13	1.000

Minimum 1.000



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Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2- Channel outl	125.35	2202.47	0.057
N1- Channel inle	125.35	2202.47	0.057
Shell inlet	465.64	2294.24	0.203
Shell outlet	465.64	2294.24	0.203

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	22.105 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	12.594 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.351 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	18.704 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	13.853 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	13.853 kgf/cm ²
App 27-4	= M.A.W.P.	17.004 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.945 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.312 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.983 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	10.107 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	8.044 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8.044 kgf/cm ²
App 27-4	= M.A.W.P.	9.188 kgf/cm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 9.688 • 1.0
 = 12.594 kgf/cm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 5.625 • 1.0
 = 7.312 kgf/cm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Chnl. head	335.9	2386.0	0.141	7.39
Chnl. rear	343.1	2386.0	0.144	7.39
Shell	406.8	2386.0	0.171	12.67
shell head	723.6	2386.0	0.303	12.67

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2- Channel outl	Nozzle	1203.20	1203.20	1.000
N2- Channel outl	Pad	1407.13	1407.13	1.000
N1- Channel inle	Nozzle	1203.20	1203.20	1.000
N1- Channel inle	Pad	1407.13	1407.13	1.000
Shell inlet	Nozzle	1407.13	1407.13	1.000
Shell outlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



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Chnl. head	1407.13	1407.13	1.000
Chnl. rear	1407.13	1407.13	1.000
Chnl. rear flange-2	1407.13	1407.13	1.000
Shell flange	1407.13	1407.13	1.000
Shell	1407.13	1407.13	1.000
shell head	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2- Channel outl	77.06	2202.47	0.035
N1- Channel inle	77.06	2202.47	0.035
Shell inlet	266.02	2294.24	0.116
Shell outlet	266.02	2294.24	0.116

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		12.59 kgf/cm ²
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	800.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		7.31 kgf/cm ²
Channel Thickness	tc	12.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	800.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	302
Tube Wall Thickness	et	2.5200 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6096.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5996.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1835.39 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.5200 mm
Groove Weld Leg	ag	2.5200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	379.350 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:



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Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	100.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1020.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	866.00 mm
Step 1 Depth on the Tubeside	ht1	8.00 mm
Step 2 Diameter on the Tubeside	dt2	797.00 mm
Step 2 Depth on the Tubeside	ht2	8.00 mm
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	866.00 mm
Step 1 Depth on the Shellside	hs1	8.00 mm
Step 2 Diameter on the Shellside	ds2	797.00 mm
Step 2 Depth on the Shellside	hs2	8.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		866.00 mm

Note: Tubesheet diameter is now: 866.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	845.151 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	1020.000 mm
Flange Inside Diameter	B	800.000 mm
Flange Face Outside Diameter	Fod	869.000 mm
Flange Face Inside Diameter	Fid	800.000 mm
Gasket Outside Diameter	Go	858.000 mm
Gasket Inside Diameter	Gi	832.000 mm
Small end Hub thk.	g0	16.0000 mm
Large end Hub thk.	g1	24.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	703.07 kgf/cm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	940.000 mm
Nominal Bolt Diameter	dB	25.4000 mm
Thread Series	UNC Thread Series	
Number of Bolts	n	24
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	1753.82 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82 kgf/cm ²
Weld between Flange and Shell/Channel		0.0000 mm

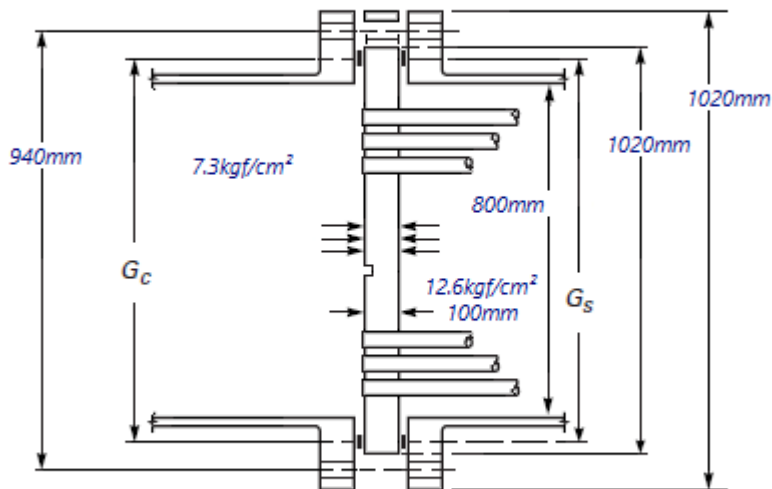


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Alternate Flange Operating Bolt Load, Wm1 83535.97 kgf
 Alternate Flange Seating Bolt Load, Wm2 119929.54 kgf
 Alternate Flange Design Bolt Load, W 134779.06 kgf

Tubesheet Integral with None
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 68.0000 mm
 Is Bolt Load Transferred to the Tubesheet No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubesheet Design Temp.	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$



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$$= (7.31 \cdot (25.4/2-0)) / (942.17 \cdot 1.0 + 0.4 \cdot 7.31)$$

$$= 0.0983 + 0.0000 = 0.0983 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.5200	25.40	6375.35	10.08	50.0000	0.0108275	977.27

$$\text{EMAWP} = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 977.2673) / (3 \cdot 10.0794) = 129.2763 \text{ kgf/cm}^2$$

Results for Reqd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5168	25.40	6375.35	49.15	50.0000	0.0004554	464.23

$$\text{EMAWP} = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 464.2332) / (3 \cdot 49.1486) = 12.594 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5168 mm
Allowable Internal Pressure at Corroded thickness	203.07 kgf/cm²
Required Internal Design Pressure	7.31 kgf/cm²
Allowable External Pressure at Corroded thickness	129.28 kgf/cm²
Required External Design Pressure	12.59 kgf/cm²
Required Thickness due to Shell Side pressure	0.5168 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	845.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 379.35 + 25.4$$

$$= 784.100 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Gs / Do \text{ (Configurations d, e, f)}$$

$$= 845.1556 / 784.1$$

$$= 1.0779$$

Ratio [pc]:

$$= Gc / Do \text{ (Configurations d)}$$



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$$= 845.1509/784.1$$

$$= 1.0779$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/100.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t(EtT/E) \cdot (StT/S) \cdot (p); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.52 (2067025/2067025) \cdot (942/1407) \cdot (0); 25.4 - 2 \cdot 2.52)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.75 / \sqrt{ (1 - 4 \cdot \min(0 ; 4 \cdot 784.1 \cdot 31.75) / (\pi \cdot 784.1^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

$$h/p = 3.149606 ; \mu^* = 0.200000 ; E^* = 321110 \text{ kgf/cm}^2$$

$$E^*/E = 0.155349 ; v^* = 0.458572$$

Note: As h/p (3.150) is > 2, data values for h/p = 2 were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 866.0/784.1$$

$$= 1.1045$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.46) / 321110 \cdot (2067025 \cdot \ln(1.1))$$

$$= 0.3463$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 784.1^2/16 \cdot [(1.078 - 1) \cdot (1.078^2 + 1) \cdot 12.594 - (1.078 - 1) \cdot (1.078^2 + 1) \cdot 7.312]$$

$$= 34165.9375 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$$

$$= 34165.9 + 83536.0 \cdot (845.151 - 845.151) / (2 \cdot \pi \cdot 784.1)$$

$$= 34165.9375 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Note: W* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((34165.94) - 784.1^2/32 \cdot 0.346 (12.59 - 7.31)) / (1 + 0.35)$$



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$$= -719.6982 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\ &= -719.7 + 784.1^2/64 \cdot (3 + 0.459) \cdot (12.59 - 7.31) \\ &= 174753.9844 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|-719.698|; |174753.984|) \\ &= 174753.9844 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 174753.984 / (0.2 \cdot (100.0 - 8.0)^2) \\ &= 619.4022 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta_T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 2549.2 \\ &= 4843.40 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 174806.891 / (0.2 \cdot (40.9101 - 8.0)^2) \\ &= 4841.9492 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 40.9101 + 0 + 0 \\ &= 40.9101 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(40.9101; 2.6719) \\ &= 40.9101 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |12.59 - 7.31| \\ &= 5.281 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y \\ &= 4 \cdot 0.2 \cdot 100.0 / 784.1 \cdot 0.8 \cdot 0.95 \cdot 2549.156 \\ &= 197.664 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta_T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.2) \cdot (784.1 / 1937.359) \cdot |12.59 - 7.31| + 0 \\ &= 2.6719 \text{ mm} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.52) 2.52 \\ &= 1.811 \text{ cm}^2 \end{aligned}$$



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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.52 (25.4 - 2.52) 1743.623$$

$$= 3158.337 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 1743.62$$

$$= 3158.3367 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 1743.623 / 1743.623$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 1743.62$$

$$= 2056.5935 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d) - 0.75 \cdot d_o} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.52 \cdot (25.4 - 2.52) \cdot 1.0 \cdot 1.0) - 0.75 \cdot 25.4} \right)$$

$$= 3.111 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.5557 mm

Minimum Required Groove Weld Leg [agr]: 1.5557 mm

Tube-Tubesheet Jt allowable, 3158.34 is $\geq$ tube strength 3158.34 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	619.4 $\leq$	4843.4 kgf/cm ²	Ok
Tubesheet shear stress	51.8 $\leq$	1937.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	40.910	100.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.556	2.520	Ok
Tube-Tubesheet Groove Weld Leg :	1.556	2.520	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
		Bend	Allwd	Shear	Allwd			
D1uc	46.766	1.351	858	4843	72	1937	Ps+Pt D1	Ok
D2uc	58.826	2.327	1477	4843	123	1937	Ps+Pt D2	Ok
D3uc	40.910	0.976	619	4843	52	1937	Ps+Pt D3	Ok



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Max: 58.8258 2.327 mm 0.305 0.064 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
Because there was no net pressure differential across the tubesheet due to
the specified pressures psd,min and ptd,min; load case D4 was not processed as
the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	58.826	100.000	Ok
Tubesheet thickness flanged extension :	2.327	68.000	Ok
Tube thickness :	0.517	2.520	Ok
Tube-tubesheet fillet weld leg :	1.556	2.520	Ok
Tube-tubesheet groove weld leg :	1.556	2.520	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	824	8.8	0.0014833	983.97
20	30	81.6667	824	8.8	0.0015106	988.796
30	40	No Calc	...	76.8	No Calc	No Calc
40	50	No Calc	...	60	No Calc	No Calc
50	60	641.667	832	12.8	0.00030782	313.804
60	70	No Calc	820.4	7	0.0011851	921.271

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	12	No Calc	...	11.676
20	30	12	No Calc	...	14.0799
30	40	80	75.7428	...	No Calc
40	50	60	54.0258	...	No Calc
50	60	16	No Calc	...	6.437
60	70	10.2	No Calc	...	8.73408

Minimum

6

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	81.6667	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	641.667	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 54 of 170
 Element and Detail Weights: Step: 13 9:54am Aug 12,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	85.2559	92153.6	62.521	94178.7	8.52559
20	30	166.076	351859	122.269	357511	16.6076
30	40	214.964	52970.6	208.456	53116	21.4964
40	50	175.319	46244.3	175.319	46244.3	17.5319
50	60	2002.73	2170857	1608.47	2221727	200.273
60	70	85.2559	92153.6	62.521	94178.7	8.52559
Total		2729	2806239.00	2239	2866957.00	272



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	92.1121	-6.66667	...	...	Cooling water
20	Liqd	351.701	35	...	...	Cooling water
20	Nozl	28.9218	30	48.4137	...	N2- Channel outl
20	Nozl	28.9218	30	-48.4137	...	N1- Channel inle
20	Wght	69.9988	...	...	...	Partition plate
50	Sadl	102.069	126	51.6	...	Left Saddle
50	Sadl	102.069	504	51.6	...	Right saddle
50	Liqd	10.8544	315	...	...	Hydrogen
50	Insl	50.9439	315	...	...	PP
50	Nozl	200.23	50	60.32	...	Shell inlet
50	Nozl	200.23	590	-60.32	...	Shell outlet
60	Liqd	0.46077	11.6667	...	...	Hydrogen
60	Insl	3.37981	11.25	...	...	Hot insulation
30	FTsh	566.174	15.25	...	...	Fixed tubesheet
30	Tube	3028.56	315.05	...	...	Tubes
40	Bafl	11.9407	90.481	...	...	Baffles
40	Bafl	11.9407	190.941	...	...	Baffles
40	Bafl	11.9407	291.401	...	...	Baffles
40	Bafl	11.9407	391.861	...	...	Baffles
40	Bafl	11.9407	492.321	...	...	Baffles
40	Bafl	11.9407	592.781	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	204.1
Liquid	455.1
Insulation	54.3
Nozzles	458.3
Weights	70.0
Exchanger Components	3666.4
Liquid in Tubes	657.7
Sum of the Detail Weights	5565.9 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
--	------------	-----------	----------	---------	-------	-----------



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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 55 of 170
 Element and Detail Weights: Step: 13 9:54am Aug 12,2026

Main Elements	3002.6	3002.6	3002.6	3002.6	3002.6	3002.6
Saddles	204.1	204.1	204.1	204.1	204.1	204.1
Nozzles	458.3	458.3	458.3	458.3	458.3	458.3
Wld Weights	70.0	70.0	70.0	70.0	70.0	70.0
Exchanger	3666.4	3666.4	3666.4	3666.4	3666.4	3666.4
Insulation	...	...	...	...	54.3	54.3 *
Ope. Liquid	...	...	...	...	...	455.1
Tube Ope Lqd	...	...	...	...	...	657.7
Test Liquid	...	2805.0	...	...	...	...
Tube Tst Lqd	...	657.7	...	...	...	...
Totals	7401.4	10864.0	7401.4	7401.4	7455.7	8568.5

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	7401.4 kg
Shop Test Wt.	- Fabricated weight + water (full)	10864.0 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	7401.4 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	7401.4 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	7455.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	8568.5 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	8029.4 kg
Field Test Wt.	- Empty weight + test medium (full)	10440.0 kg

Exchanger Tube Data

Volume of Exchanger tubes :	657963.2 cm ³
Weight of Ope Liq in tubes :	657.7 kg
Weight of Water in tubes :	657.7 kg

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	8654.34	8194.14
20	30	18120.7	17592.9
30	40	6373.29	2613.81
40	50	5963.12	2312.21
50	60	164670	158336
60	70	8654.34	8194.14

Total	212435.5	197243.5 cm ²
Total	228.7	212.3 ft ²



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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 56 of 170
 Nozzle Flange MAWP: Step: 14 9:54am Aug 12,2026

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2- Channel ou	18.87	19.99	71	150	GR 1.1	32.336	9.188	18.875	-4.024
N1- Channel in	18.87	19.99	71	150	GR 1.1	32.336	9.188	18.875	-4.024
Shell inlet	17.00	19.99	127	150	GR 1.1	19.827	17.004	17.004	5.679
Shell outlet	17.00	19.99	127	150	GR 1.1	19.827	17.004	17.004	5.679

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 17.004 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 19.987 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 9.188 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 19.987 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm²	Maximum Pressure Pmax kgf/cm²	Pass Fail
N2- Channel outl	196.088	1.20	459	408	9.188	9.188	Pass
N1- Channel inle	196.088	1.20	459	408	9.188	9.188	Pass
Shell inlet	442.976	1.20	3262	1101	17.004	17.004	Pass
Shell outlet	442.976	1.20	3262	1101	17.004	17.004	Pass

Evaluating Flange on Nozzle [N2- Channel outl]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(459) + 4 \cdot \text{abs}(408) 196.088 \leq \pi \cdot 196.088^3 [(18.875 - 9.188) + 1.200 \cdot 18.875]$$

$$= 0.76593E+04 \leq 0.76593E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2- Channel outl [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.875, 1.200 \cdot 18.875 + 18.875 - 7659.3 / (\pi \cdot 196.088^3))$$

$$= \min(18.875, 9.188)$$

$$= 9.188 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N1- Channel inle]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(459) + 4 \cdot \text{abs}(408) 196.088 \leq \pi \cdot 196.088^3 [(18.875 - 9.188) + 1.200 \cdot 18.875]$$

$$= 0.76593E+04 \leq 0.76593E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1- Channel inle [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.875, 1.200 \cdot 18.875 + 18.875 - 7659.3 / (\pi \cdot 196.088^3))$$

$$= \min(18.875, 9.188)$$

$$= 9.188 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [Shell inlet]:



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Nozzle Flange MAWP: Step: 14 9:54am Aug 12,2026

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(3262) + 4 \cdot \text{abs}(1101) 442.976 \leq \pi \cdot 442.976^3 [(17.004 - 17.004) + 1.200 \cdot 17.004] \\
 &= 0.54143E+05 \leq 0.55720E+05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell inlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(17.004, 1.200 \cdot 17.004 + 17.004 - 54142.5 / (\pi \cdot 442.976^3)) \\
 &= \min(17.004, 17.581) \\
 &= 17.004 \text{ kgf/cm}^2
 \end{aligned}$$

Evaluating Flange on Nozzle [Shell outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(3262) + 4 \cdot \text{abs}(1101) 442.976 \leq \pi \cdot 442.976^3 [(17.004 - 17.004) + 1.200 \cdot 17.004] \\
 &= 0.54143E+05 \leq 0.55720E+05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(17.004, 1.200 \cdot 17.004 + 17.004 - 54142.5 / (\pi \cdot 442.976^3)) \\
 &= \min(17.004, 17.581) \\
 &= 17.004 \text{ kgf/cm}^2
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 58 of 170
 Wind Load Calculation: Step: 15 9:54am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020		
Basic Wind Speed for IS-875	180	km/hr	
Wind Zone Number	5		
Base Elevation	30	cm	
Percent Wind for Hydrotest	33.0		
Risk Factor	[k1]	1.0	
Terrain Category		2	
Equipment Class		Class A	
Topography Factor	[k3]	1.0	
Importance Factor	[k4]	1.15	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Use Gust Response Factor (Dynamic Analysis)		No	

Calculated Values:

Vessel Length to Diameter ratio: 8.847

Design Wind Speed [Vz]:
 = Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)
 = 50.0 • 1.0 • 1.0 • 1.0 • 1.15
 = 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:
 = max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)
 = max(0.7 • 1983.499, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)
 = 1983.5 N/m²

Element Wind Load (operating condition):
 = Wind Area • Cf • pd
 = 0.2 • 0.877 • 1983.499
 = 324.6 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	0.96	1.0000	1.0000	1.0000	57.50	0.8769
20	0.96	1.0000	1.0000	1.0000	57.50	0.8769
30	0.96	1.0000	1.0000	1.0000	57.50	0.8769
40	0.96	1.0000	1.0000	1.0000	57.50	0.8769
50	0.96	1.0000	1.0000	1.0000	57.50	0.8769
60	0.96	1.0000	1.0000	1.0000	57.50	0.8769

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm²	Elevation cm	Pressure kgf/m²	Force kgf
Partition pla	0.00	96.00	202.34	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
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Wind Load Calculation: Step: 15 9:54am Aug 12,2026

10	20	96	98.88	1866.4	202.344	36.4163
20	30	96	98.88	6921.6	202.344	135.051
30	40	96	96	940.8	202.344	18.3565
40	50	96	96	806.4	202.344	15.7341
50	60	96	105.84	66679.2	202.344	1301.01
60	70	96	104.448	2130.09	202.344	41.5614

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 FileName : 403-E-001ABC_Hydrogen Compressor 1st stage inter Page 60 of 170
 Earthquake Load Calculation: Step: 16 9:54am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.0) \cdot 1.0$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:

$= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 1045.55 \cdot 0.0939 \cdot 1.0$
 $= 98.1 \text{ kgf}$

Vertical load per 8.4:

$= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0939 \cdot 8364 \cdot 1.0$



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Earthquake Load Calculation: Step: 16 9:54am Aug 12,2026

= 523.4 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	40	1045.55	98.1308
20	30	40	1045.55	98.1308
30	40	40	1045.55	98.1308
40	50	40	1045.55	98.1308
50	Sad1	40	1045.55	98.1308
Sad1	60	40	1045.55	98.1308
50	60	40	1045.55	98.1308
60	70	40	1045.55	98.1308

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 17 9:54am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	418.650 cm
Center of Gravity of the Liquid	40.300 cm
Center of Gravity of the Insulation	438.948 cm
Center of Gravity of the Nozzles	374.597 cm
Center of Gravity of the Added Weights (Operating)	5.000 cm
Center of Gravity of the Added Weights (Empty)	5.000 cm
Center of Gravity of the Tubesheet(s)	90.250 cm
Center of Gravity of the Tubes	390.050 cm
Center of Gravity of the Baffles	436.881 cm
Center of Gravity of Bare Shell New and Cold	345.388 cm
Center of Gravity of Bare Shell Corroded	338.748 cm
Vessel CG in the Operating Condition	330.583 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	345.642 cm
Vessel CG in the Test Condition	351.079 cm

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 Saddle Calcs: Bundle Pull New: Step: 18 9:54am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

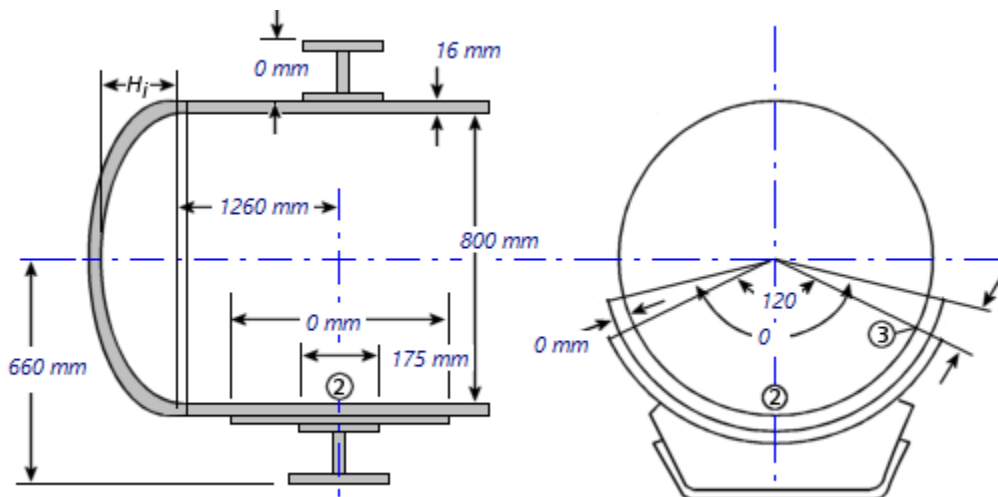
Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{408.0 \cdot 16.0}, 2 \cdot 1260.0)$
 $= 301.0418 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	408.00	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	643.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	175.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		3701.75	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		5028.15	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual	Allowable
	kgf/cm ²	kgf/cm ²



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 Saddle Calcs: Bundle Pull New: Step: 18 9:54am Aug 12,2026

Long. Stress at Top of Midspan	-21.72	-2386.01
Long. Stress at Top of Midspan	21.72	1407.13
Long. Stress at Bottom of Midspan	21.72	1407.13
Long. Stress at Top of Saddles	131.79	1407.13
Long. Stress at Bottom of Saddles	-73.05	-2386.01
Long. Stress at Bottom of Saddles	73.05	1407.13
Tangential Shear in Shell	54.85	1125.71
Circ. Stress at Horn of Saddle	181.81	2110.70
Circ. Compressive Stress in Shell	7.94	1407.13

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 3702 • 1.5 • 660.0/3780.0007
 = 969.5 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 4059 + max(970, 0, 0, 0)
 = 5028.1 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5130.21	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	5552.63	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 720.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-5028 \cdot 126.0 [1 - (1 - 126.0/643.4 + (40.8^2 - 0.0^2)/(2 \cdot 126.0 \cdot 643.4))/(1 + (4 \cdot 0.0)/(3 \cdot 643.4))]$
 = -1175.7 kgf-m.

Moment per Equation 4.15.2 [M2]:
 = $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $5028 \cdot 643/4 ((1 + 2(41^2 - 0^2)/(643^2))/(1 + (4 \cdot 0)/(3 \cdot 643)) - 4 \cdot 126/643)$
 = 1817.4 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 = $0.0 \cdot 408.0/(2 \cdot 16.0) - 1817.4/(\pi \cdot 408.0^2 \cdot 16.0)$
 = -21.72 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0.0 \cdot 408.0/(2 \cdot 16.0) + 1817.4/(\pi \cdot 408.0^2 \cdot 16.0)$
 = 21.72 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:



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 Saddle Calcs: Bundle Pull New: Step: 18 9:54am Aug 12,2026

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 408.0 / (2 \cdot 16.0) - 1175.7 / (0.1066 \cdot \pi \cdot 408.0^2 \cdot 16.0)$$

$$= 131.79 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 408.0 / (2 \cdot 16.0) + 1175.7 / (0.1923 \cdot \pi \cdot 408.0^2 \cdot 16.0)$$

$$= -73.05 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 5028(643.4 - 2 \cdot 126.0) / (643.4 + (4 \cdot 0.0 / 3))$$

$$= 3058.8 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3058.78 / (408.0 \cdot 16.0)$$

$$= 54.85 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(408.0 \cdot 16.0)}$$

$$= 63.021 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 5028 \cdot 0.1 / (16.0(175.0 + 63.02 + 63.02))$$

$$= -7.94 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -5028 / (4 \cdot 16.0(175.0 + 63.021 + 63.021)) - 3 \cdot 0.0529 \cdot 5028 / (2 \cdot 16.0^2)$$

$$= -181.81 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{(408.0 \cdot 16.0)}, 2 \cdot 126.0)$$

$$= 301.04 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(400.0 \cdot 16.0)} + 225.0)^3) / 12$$

$$= 11.940 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 11.94 + 55.968(83.304 - 8.0)^2$$

$$= 3185.744 \text{ cm}^4$$

B

D

Y

A

AY

I + AD⁽²⁾



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	mm	mm	mm	cm ²	mm ³	cm ⁴
Shell	349.8	16.0	8.0	56.0	44774.4	0.319E+04
Wearplate	225.0	12.0	22.0	27.0	59400.0	0.102E+04
Web	16.0	216.0	136.0	34.6	470016.1	0.230E+04
BasePlate	150.0	16.0	252.0	24.0	604800.0	0.684E+04
Totals	...	...	...	141.5	1178990.6	0.133E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1178991/141.528$$

$$= 83.304 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 5028.146$$

$$= 1023.3367 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.9605 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 400.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 329.203 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 336.8856 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 21.0341 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(5028 + 102) \cdot 150.0) / (2 \cdot 720.0 \cdot 1767.415)}$$

$$= 9.524 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 150.0/2$$

$$= 277.994 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



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$$= 5130.215 / (720.0 \cdot 150.0)$$

$$= 4.750 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 278.0 \cdot 4.75$$

$$= 1320.528 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 29.653$$

$$= 55.093 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1320.5 / 55.0927$$

$$= 23.969 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.110E+04
Web	185.3	16.0	8.0	29.7	23722.2	490.
Totals	...	...	...	55.1	266674.2	0.159E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 266674.156 / 55.093$$

$$= 48.405 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(48.405, 175.0 - 48.405)$$

$$= 126.595 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1590.6 / 55.093)}$$

$$= 53.732 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 660.0 - \cos(120.0 / 2) (400.0 + 16.0 + 12.0) - 16.0$$

$$= 430.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 720.0 - 50.8$$

$$= 669.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$



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$$= (741.318 - 669.2) / 2$$

$$= 36.059 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{ (L_v^2 + D_d^2) }$$

$$= \sqrt{ (430.0^2 + 36.059^2) }$$

$$= 431.509 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2038900 / 2651.1) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 431.509 / 53.732$$

$$= 8.031$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 5552.6 / (2 \cdot 720.0) \cdot 370.659 \cdot 431.51 / 2$$

$$= 308.369 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.0307 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.03)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (8.03) / (8 \cdot 123.21) - (8.03^3) / (8 \cdot 123.21^3))$$

$$= 1564 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 23.97 / 1564.39 + (308.37 \cdot 126.595 / 15906263) / 1767.42$$

$$= 0.154$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.149E+04
Web	370.7	16.0	8.0	59.3	47444.3	422.
Totals	...	...	...	84.7	290396.3	0.191E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 290396.312 / 84.745$$

$$= 34.267 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(34.267, \text{abs}(175.0 - 34.267))$$

$$= 140.733 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 660.0 - \cos(428.0 / 2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 216.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1911.7 / 84.745) }$$

$$= 47.495 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 216.0 / 47.495$$

$$= 4.548$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 5552.627 / (2 \cdot 720.0)$$

$$= 3.856 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.856 \cdot 370.659 \cdot 216.0$$

$$= 308.720 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 308.72 / 135835.672$$

$$= 227.275 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.5479 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.55)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (4.55) / (8 \cdot 123.21) - (4.55)^3 / (8 \cdot 123.21^3))$$

$$= 1577 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 34.44 / 1576.5 + (308.72 \cdot 140.733 / 1911.656) / 1767.42$$

$$= 0.150$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	5130.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	5552.6 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	F1	5552.6 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5552.6 / (4.235 \cdot 2)$$

$$= 655.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (4.235 \cdot 2)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (Q_L - Q_0) / (Stba \cdot R_{nbolts})$$

$$= (5552.63 - 5130.21) / (1753.82 \cdot 2.0)$$

$$= 0.1204 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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$$= 5552.63 / (1052.29 \cdot 2.0)$$
$$= 2.6384 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $660.0 \cdot 0.0 + 0.0$
= 0.00 kgf-m.

Eccentricity (e):
= $Rmom / QO$
= $0.0 / 5130.21$
= 0.00 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.1204, 2.6384, 0.0]$
= 2.6384 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

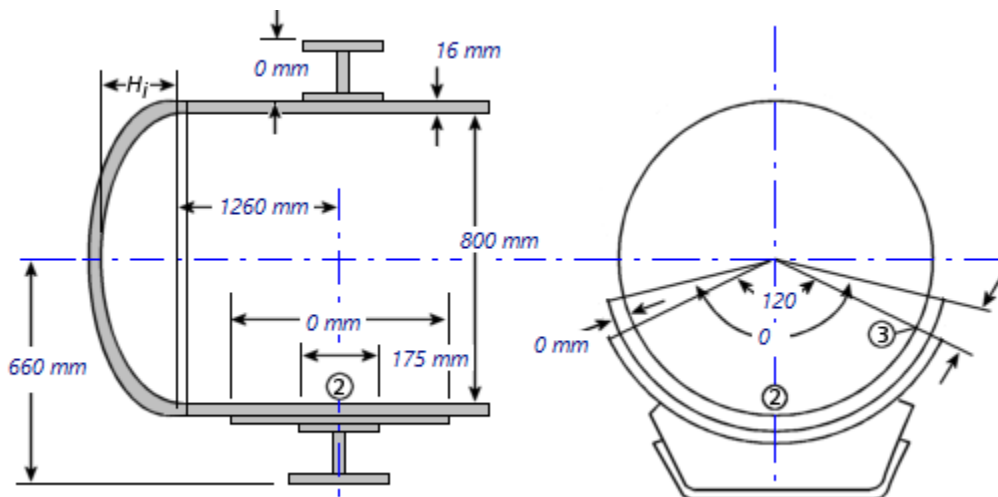
Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{409.6 \cdot 12.8}, 2 \cdot 1260.0)$
 $= 287.9561 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	409.60	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	643.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	175.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		3701.75	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		4758.62	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual	Allowable
	kgf/cm ²	kgf/cm ²



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Long. Stress at Top of Midspan	-25.50	-2386.01
Long. Stress at Top of Midspan	25.50	1407.13
Long. Stress at Bottom of Midspan	25.50	1407.13
Long. Stress at Top of Saddles	154.63	1407.13
Long. Stress at Bottom of Saddles	-85.70	-2386.01
Long. Stress at Bottom of Saddles	85.70	1407.13
Tangential Shear in Shell	64.64	1125.71
Circ. Stress at Horn of Saddle	262.53	2110.70
Circ. Compressive Stress in Shell	9.82	1407.13

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 3702 • 1.5 • 660.0/3780.0007
 = 969.5 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 3789 + max(970, 0, 0, 0)
 = 4758.6 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4860.69	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	5552.63	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 720.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-4759 \cdot 126.0 [1 - (1 - 126.0/643.4 + (40.96^2 - 0.0^2)/(2 \cdot 126.0 \cdot 643.4))/(1 + (4 \cdot 0.0)/(3 \cdot 643.4))]$
 = -1112.2 kgf-m.

Moment per Equation 4.15.2 [M2]:
 = $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $4759 \cdot 643/4 ((1 + 2(41^2 - 0^2)/(643^2))/(1 + (4 \cdot 0)/(3 \cdot 643)) - 4 \cdot 126/643)$
 = 1720.4 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 = $0.0 \cdot 409.6/(2 \cdot 12.8) - 1720.4/(\pi \cdot 409.6^2 \cdot 12.8)$
 = -25.50 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0.0 \cdot 409.6/(2 \cdot 12.8) + 1720.4/(\pi \cdot 409.6^2 \cdot 12.8)$
 = 25.50 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:



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$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 409.6 / (2 \cdot 12.8) - 1112.2 / (0.1066 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 154.63 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 409.6 / (2 \cdot 12.8) + 1112.2 / (0.1923 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= -85.70 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 4759(643.4 - 2 \cdot 126.0) / (643.4 + (4 \cdot 0.0 / 3))$$

$$= 2894.8 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2894.81 / (409.6 \cdot 12.8)$$

$$= 64.64 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(409.6 \cdot 12.8)}$$

$$= 56.478 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4759 \cdot 0.1 / (12.8(175.0 + 56.48 + 56.48))$$

$$= -9.82 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -4759 / (4 \cdot 12.8(175.0 + 56.478 + 56.478)) - 3 \cdot 0.0529 \cdot 4759 / (2 \cdot 12.8^2)$$

$$= -262.53 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{(409.6 \cdot 12.8)}, 2 \cdot 126.0)$$

$$= 287.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 12.8)} + 225.0)^3) / 12$$

$$= 5.891 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.891 + 43.145(88.143 - 6.4)^2$$

$$= 2888.831 \text{ cm}^4$$



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	mm	mm	mm	cm ²	mm ³	cm ⁴
Shell	337.1	12.8	6.4	43.1	27612.8	0.289E+04
Wearplate	225.0	12.0	18.8	27.0	50760.0	0.130E+04
Web	16.0	216.0	132.8	34.6	458956.9	0.203E+04
BasePlate	150.0	16.0	248.8	24.0	597120.0	0.620E+04
Totals	...	...	...	128.7	1134449.8	0.124E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1134450 / 128.705$$

$$= 88.143 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4758.619$$

$$= 968.4822 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.3193 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.556 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 316.2643 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 22.4397 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4759 + 102) \cdot 150.0) / (2 \cdot 720.0 \cdot 1767.415)}$$

$$= 9.271 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 150.0 / 2$$

$$= 277.994 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



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$$= 4860.689 / (720.0 \cdot 150.0)$$

$$= 4.501 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 278.0 \cdot 4.5$$

$$= 1251.151 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 29.653$$

$$= 55.093 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1251.2 / 55.0927$$

$$= 22.710 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.110E+04
Web	185.3	16.0	8.0	29.7	23722.2	490.
Totals	...	...	...	55.1	266674.2	0.159E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 266674.156 / 55.093$$

$$= 48.405 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(48.405, 175.0 - 48.405)$$

$$= 126.595 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1590.6 / 55.093)}$$

$$= 53.732 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 660.0 - \cos(120.0 / 2) (400.0 + 16.0 + 12.0) - 16.0$$

$$= 430.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 720.0 - 50.8$$

$$= 669.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$



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$$= (741.318 - 669.2) / 2$$

$$= 36.059 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{ (L_v^2 + D_d^2) }$$

$$= \sqrt{ (430.0^2 + 36.059^2) }$$

$$= 431.509 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2038900 / 2651.1) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 431.509 / 53.732$$

$$= 8.031$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 5552.6 / (2 \cdot 720.0) \cdot 370.659 \cdot 431.51 / 2$$

$$= 308.369 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.0307 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.03)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (8.03) / (8 \cdot 123.21) - (8.03^3) / (8 \cdot 123.21^3))$$

$$= 1564 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 22.71 / 1564.39 + (308.37 \cdot 126.595 / 15906263) / 1767.42$$

$$= 0.153$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.149E+04
Web	370.7	16.0	8.0	59.3	47444.3	422.
Totals	...	...	...	84.7	290396.3	0.191E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 290396.312 / 84.745$$

$$= 34.267 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(34.267, \text{abs}(175.0 - 34.267))$$

$$= 140.733 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 660.0 - \cos(428.0 / 2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 216.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1911.7 / 84.745) }$$

$$= 47.495 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 216.0 / 47.495$$

$$= 4.548$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 5552.627 / (2 \cdot 720.0)$$

$$= 3.856 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.856 \cdot 370.659 \cdot 216.0$$

$$= 308.720 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 308.72 / 135835.672$$

$$= 227.275 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (4.5479 < 123.2106) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.55)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (4.55) / (8 \cdot 123.21) - (4.55^3) / (8 \cdot 123.21^3))$$

$$= 1577 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 32.63 / 1576.5 + (308.72 \cdot 140.733 / 1911.656) / 1767.42$$

$$= 0.149$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	4860.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	5552.6 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	F1	5552.6 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5552.6 / (4.235 \cdot 2)$$

$$= 655.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (4.235 \cdot 2)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (Q_L - Q_0) / (Stba \cdot R_{nbolts})$$

$$= (5552.63 - 4860.69) / (1753.82 \cdot 2.0)$$

$$= 0.1973 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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$$= 5552.63 / (1052.29 \cdot 2.0)$$
$$= 2.6384 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $660.0 \cdot 0.0 + 0.0$
= 0.00 kgf-m.

Eccentricity (e):
= $Rmom / QO$
= $0.0 / 4860.69$
= 0.00 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.1973, 2.6384, 0.0]$
= 2.6384 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

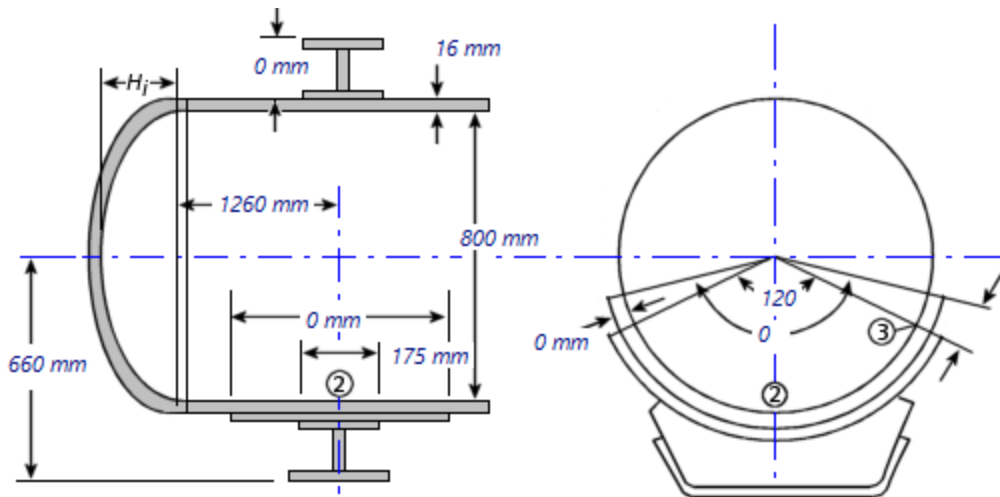
Horizontal Vessel Stress Calculations : Operating Case

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{409.6 \cdot 12.8}, 2 \cdot 1260.0)$
 $= 287.9561 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	409.60	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	643.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	175.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		8504.46	kgf
Pressure used in Analysis	P	9.688	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	109.43	1407.13
Long. Stress at Bottom of Midspan	200.58	1407.13
Long. Stress at Top of Saddles	431.35	1407.13
Long. Stress at Bottom of Saddles	1.84	1407.13



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Tangential Shear in Shell	115.52	1125.71
Circ. Stress at Horn of Saddle	469.19	1758.92
Circ. Compressive Stress in Shell	17.54	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1548.1/2 + 0) \cdot 660.0/720.0$
 $= 2128.7\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(174.25, 0) \cdot 660.0/3780.0007$
 $= 30.4\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(785.05, 0) \cdot 660.0/3780.0007$
 $= 137.1\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(785/2 + 0) \cdot 660.0/720.0$
 $= 1079.4\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 6376 + \max(30, 2129, 137, 1079)$
 $= 8504.5\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1984.302(0.0 + \pi/4(0.832 \cdot 1.2)^2)$
 $= 1708.845\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8606.53	kgf
Transverse Shear Load	774.07	kgf
Longitudinal Shear Load	785.05	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 720.000\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)]$
 $= -8504 \cdot 126.0 [1 - (1 - 126.0/643.4 + (40.96^2 - 0.0^2) / (2 \cdot 126.0 \cdot 643.4)) / (1 + (4 \cdot 0.0) / (3 \cdot 643.4))]$
 $= -1987.6\ kgf \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 8504 \cdot 643 / 4 ((1 + 2(41^2 - 0^2)/(643^2)) / (1 + (4 \cdot 0) /$



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$$(3 \cdot 643) - 4 \cdot 126 / 643 \\ = 3074.7 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 9.688 \cdot 409.6 / (2 \cdot 12.8) - 3074.7 / (\pi \cdot 409.6^2 \cdot 12.8) \\ = 109.43 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 9.688 \cdot 409.6 / (2 \cdot 12.8) + 3074.7 / (\pi \cdot 409.6^2 \cdot 12.8) \\ = 200.58 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 9.688 \cdot 409.6 / (2 \cdot 12.8) - 1987.6 / (0.1066 \cdot \pi \cdot 409.6^2 \cdot 12.8) \\ = 431.35 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 9.688 \cdot 409.6 / (2 \cdot 12.8) + 1987.6 / (0.1923 \cdot \pi \cdot 409.6^2 \cdot 12.8) \\ = 1.84 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 8504(643.4 - 2 \cdot 126.0) / (643.4 + (4 \cdot 0.0 / 3)) \\ = 5173.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 5173.53 / (409.6 \cdot 12.8) \\ = 115.52 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{409.6 \cdot 12.8} \\ = 56.478 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2)) \\ = -0.7603 \cdot 8504 \cdot 0.1 / (12.8(175.0 + 56.48 + 56.48)) \\ = -17.54 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2) \\ = -8504 / (4 \cdot 12.8(175.0 + 56.478 + 56.478)) - \\ 3 \cdot 0.0529 \cdot 8504 / (2 \cdot 12.8^2) \\ = -469.19 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(175.0 + 1.56 \cdot \sqrt{409.6 \cdot 12.8}, 2 \cdot 126.0) \\ = 287.96 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 3780.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000012 \cdot 3780.001(127.0 - 21.1) \\ = 4.914 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm



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Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(403.2 \cdot 12.8)} + 225.0)12.8^3)/12$
 $= 5.891 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 5.891 + 43.145(88.143 - 6.4)^2$
 $= 2888.831 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	337.1	12.8	6.4	43.1	27612.8	0.289E+04
Wearplate	225.0	12.0	18.8	27.0	50760.0	0.130E+04
Web	16.0	216.0	132.8	34.6	458956.9	0.203E+04
BasePlate	150.0	16.0	248.8	24.0	597120.0	0.620E+04
Totals	...	...	...	128.7	1134449.8	0.124E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 1134450/128.705$
 $= 88.143 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120.0/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 8504.461$
 $= 1730.8423 \text{ kgf}$

Tension Stress, St = (Fh/As) = 20.2296 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1602.9094 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 660.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$
 $= 326.556 \text{ mm}$

Bending Moment, M = Fh * d = 565.2181 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 40.1035 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1781.0104 kgf/cm²

Minimum Thickness of Baseplate per Moss:



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$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(8504 + 102)150.0 / (2 \cdot 720.0 \cdot 1781.01)}$$

$$= 12.289 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 150.0 / 2$$

$$= 277.994 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 8606.531 / (720.0 \cdot 150.0)$$

$$= 7.969 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 278.0 \cdot 7.97$$

$$= 2215.338 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 29.653$$

$$= 55.093 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 2215.3 / 55.0927$$

$$= 40.211 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.110E+04
Web	185.3	16.0	8.0	29.7	23722.2	490.
Totals	...	...	...	55.1	266674.2	0.159E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 266674.156 / 55.093$$

$$= 48.405 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(48.405, 175.0 - 48.405)$$

$$= 126.595 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(1590.6/55.093)}$
 $= 53.732 \text{ mm}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 660.0 - \cos(120.0/2) (400.0 + 16.0 + 12.0) - 16.0$
 $= 430.000 \text{ mm}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0/2) (400.0 + 16.0 + 12.0)$
 $= 741.318 \text{ mm}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 720.0 - 50.8$
 $= 669.200 \text{ mm}$

Length of Outer Rib, step 4, [Dd]:
 $= (Lw - Bd) / 2$
 $= (741.318 - 669.2) / 2$
 $= 36.059 \text{ mm}$

Length of Outer Rib [L]:
 $= \sqrt{L_v^2 + D_d^2}$
 $= \sqrt{430.0^2 + 36.059^2}$
 $= 431.509 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2671.5)}$
 $= 122.739$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 431.509 / 53.732$
 $= 8.031$

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 785.0 / (2 \cdot 720.0) \cdot 370.659 \cdot 431.51 / 2$
 $= 43.598 \text{ kgf-m.}$

Compressive Allowable, $KL/r < C_c$ ($8.0307 < 122.7394$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (8.03)^2 / (2 \cdot 122.74^2)) 2672 / (5/3 + 3 \cdot (8.03) / (8 \cdot 122.74) - (8.03^3) / (8 \cdot 122.74^3))$
 $= 1576 \text{ kgf/cm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 40.21 / 1576.31 + (43.6 \cdot 126.595 / 15906263) / 1781.01$
 $= 0.045$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.149E+04
Web	370.7	16.0	8.0	59.3	47444.3	422.
Totals	...	...	...	84.7	290396.3	0.191E+04



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 290396.312/84.745$$

$$= 34.267 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(34.267, \text{abs}(175.0 - 34.267))$$

$$= 140.733 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 660.0 - \cos(428.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 216.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1911.7/84.745) }$$

$$= 47.495 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 216.0/47.495$$

$$= 4.548$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 785.046 / (2 \cdot 720.0)$$

$$= 0.545 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.545 \cdot 370.659 \cdot 216.0$$

$$= 43.648 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 43.648/135835.672$$

$$= 32.133 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (4.5479 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.55)^2 / (2 \cdot 122.74^2)) 2672 /$$

$$(5/3 + 3 \cdot (4.55) / (8 \cdot 122.74) - (4.55^3) / (8 \cdot 122.74^3))$$

$$= 1589 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 57.78/1588.57 + (43.65 \cdot 140.733/1911.656) / 1781.01$$

$$= 0.054$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1399.95 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	6477.8 kgf



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Saddle Load QL (Wind/Seismic contribution)	QL	137.1 kgf
Maximum Transverse Force	Ft	774.1 kgf
Maximum Longitudinal Force	F1	785.0 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 784.2 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $\max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $785.0 / (4.235 \cdot 2)$
 = 92.7 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $774.1 / (4.235 \cdot 2)$
 = 91.4 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $785.05 / (1099.96 \cdot 2.0)$
 = 0.3569 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $660.0 \cdot 774.07 + 0.0$
 = 510.88 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $510.88 / 6477.85$
 = 78.87 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.3569, 0.0]$
 = 0.3569 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.383$, Temp. Reduction = 55 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.



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Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(175.0 + 1.56 \sqrt{ (409.6 \cdot 12.8) }, 2 \cdot 1260.0)$$

$$= 287.9561 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	409.60	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	643.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	175.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	20.67	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		4640.63	kgf
Pressure used in Analysis	P	9.688	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	134.94	1407.13
Long. Stress at Bottom of Midspan	175.07	1407.13
Long. Stress at Top of Saddles	335.05	1407.13
Long. Stress at Bottom of Saddles	55.21	1407.13
Tangential Shear in Shell	60.45	1125.71
Circ. Stress at Horn of Saddle	256.02	1758.92
Circ. Compressive Stress in Shell	9.57	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(1548.1/2 + 0) \cdot 660.0/720.0$
 $= 2128.7 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(174.25, 0) \cdot 660.0/3780.0007$
 $= 30.4 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(785.05, 0) \cdot 660.0/3780.0007$
 $= 137.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(785/2 + 0) \cdot 660.0/720.0$
 $= 1079.4 \text{ kgf}$



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Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2512 + \max(30, 2129, 137, 1079)$$

$$= 4640.6 \text{ kgf}$$

Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.302(0.0 + \pi/4(0.832 \cdot 1.2)^2)$$

$$= 1708.845 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4742.70	kgf
Transverse Shear Load	774.07	kgf
Longitudinal Shear Load	785.05	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 720.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M₁]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -4641 \cdot 126.0 [1 - (1 - 126.0/643.4 + (40.96^2 - 20.67^2)/(2 \cdot 126.0 \cdot 643.4))/(1 + (4 \cdot 20.67)/(3 \cdot 643.4))]$$

$$= -1295.0 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M₂]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 4641 \cdot 643/4((1 + 2(41^2 - 21^2)/(643^2))/(1 + (4 \cdot 21)/(3 \cdot 643)) - 4 \cdot 126/643)$$

$$= 1353.9 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 9.688 \cdot 409.6/(2 \cdot 12.8) - 1353.9/(\pi \cdot 409.6^2 \cdot 12.8)$$

$$= 134.94 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 9.688 \cdot 409.6/(2 \cdot 12.8) + 1353.9/(\pi \cdot 409.6^2 \cdot 12.8)$$

$$= 175.07 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 9.688 \cdot 409.6/(2 \cdot 12.8) - 1295.0/(0.1066 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 335.05 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 9.688 \cdot 409.6/(2 \cdot 12.8) + 1295.0/(0.1923 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 55.21 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 4641(643.4 - 2 \cdot 126.0)/(643.4 + (4 \cdot 20.67/3))$$

$$= 2707.1 \text{ kgf}$$



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 2707.08 / (409.6 \cdot 12.8)$
 $= 60.45 \text{ kgf/cm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(409.6 \cdot 12.8)}$
 $= 56.478 \text{ mm}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 4641 \cdot 0.1 / (12.8(175.0 + 56.48 + 56.48))$
 $= -9.57 \text{ kgf/cm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 $= -4641 / (4 \cdot 12.8(175.0 + 56.478 + 56.478)) -$
 $3 \cdot 0.0529 \cdot 4641 / (2 \cdot 12.8^2)$
 $= -256.02 \text{ kgf/cm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{(409.6 \cdot 12.8)}, 2 \cdot 126.0)$
 $= 287.96 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$
 $= ((1.56 \cdot \sqrt{(403.2 \cdot 12.8)} + 225.0)12.8^3) / 12$
 $= 5.891 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 5.891 + 43.145(88.143 - 6.4)^2$
 $= 2888.831 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	337.1	12.8	6.4	43.1	27612.8	0.289E+04
Wearplate	225.0	12.0	18.8	27.0	50760.0	0.130E+04
Web	16.0	216.0	132.8	34.6	458956.9	0.203E+04
BasePlate	150.0	16.0	248.8	24.0	597120.0	0.620E+04
Totals	...	...	...	128.7	1134449.8	0.124E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 1134450 / 128.705$
 $= 88.143 \text{ mm}$



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Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4640.635$$

$$= 944.4698 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.0387 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1602.9094 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.556 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 308.4229 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 21.8833 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1781.0104 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(4641 + 102) \cdot 150.0) / (2 \cdot 720.0 \cdot 1781.01))}$$

$$= 9.122 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 150.0/2$$

$$= 277.994 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4742.704 / (720.0 \cdot 150.0)$$

$$= 4.391 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 278.0 \cdot 4.39$$

$$= 1220.781 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 29.653$$

$$= 55.093 \text{ cm}^2$$



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Compressive Stress [Sc]:

= P/A_r
 = 1220.8/55.0927
 = 22.159 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.110E+04
Web	185.3	16.0	8.0	29.7	23722.2	490.
Totals	...	...	...	55.1	266674.2	0.159E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 175.0 - 16.0
 = 159.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 266674.156/55.093
 = 48.405 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(48.405, 175.0 - 48.405)
 = 126.595 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1590.6/55.093}$
 = 53.732 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 660.0 - cos(120.0/2)(400.0 + 16.0 + 12.0) - 16.0
 = 430.000 mm

Length of Outer Rib, step 2, [Lw]:

= $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 = $2 \cdot \sin(120.0/2) (400.0 + 16.0 + 12.0)$
 = 741.318 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 720.0 - 50.8
 = 669.200 mm

Length of Outer Rib, step 4, [Dd]:

= $(L_w - B_d) / 2$
 = $(741.318 - 669.2) / 2$
 = 36.059 mm

Length of Outer Rib [L]:

= $\sqrt{L_v^2 + D_d^2}$
 = $\sqrt{430.0^2 + 36.059^2}$
 = 431.509 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038900 / 2671.5}$
 = 122.739



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 431.509/53.732$$

$$= 8.031$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 785.0 / (2 \cdot 720.0) \cdot 370.659 \cdot 431.51/2$$

$$= 43.598 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (8.0307 < 122.7394) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.03)^2 / (2 \cdot 122.74^2)) 2672 /$$

$$(5/3 + 3 \cdot (8.03) / (8 \cdot 122.74) - (8.03^3) / (8 \cdot 122.74^3))$$

$$= 1576 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 22.16/1576.31 + (43.6 \cdot 126.595/15906263) / 1781.01$$

$$= 0.034$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.149E+04
Web	370.7	16.0	8.0	59.3	47444.3	422.
Totals	...	...	...	84.7	290396.3	0.191E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 290396.312/84.745$$

$$= 34.267 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(34.267, \text{abs}(175.0 - 34.267))$$

$$= 140.733 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 660.0 - \cos(428.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 216.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ (1911.7/84.745) }$$

$$= 47.495 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 216.0/47.495$$

$$= 4.548$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 785.046 / (2 \cdot 720.0)$$

$$= 0.545 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.545 \cdot 370.659 \cdot 216.0$$

$$= 43.648 \text{ kgf-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 43.648/135835.672

= 32.133 kgf/cm²

Compressive Allowable, $KL/r < C_c$ (4.5479 < 122.7394) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(4.55)²/(2 • 122.74²))2672/

(5/3+3•(4.55)/(8• 122.74)-(4.55³)/(8•122.74³)

= 1589 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 31.84/1588.57 + (43.65 • 140.733/1911.656)/1781.01

= 0.038

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1399.95 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	2614.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	137.1 kgf
Maximum Transverse Force	Ft	774.1 kgf
Maximum Longitudinal Force	F1	785.0 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 785.0/(4.235 • 2)

= 92.7 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 774.1/(4.235 • 2)

= 91.4 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= 785.05/(1099.96 • 2.0)

= 0.3569 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= 660.0 • 774.07 + 0.0

= 510.88 kgf-m.

Eccentricity (e):

= $R_{mom} / Q0$

= 510.88/2614.02



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= 195.44 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis
 = 720.0/2 - 50.0
 = 310.00 mm

Modular Ratio Of Steel/Concrete (n1):
 = ES / EC
 = 2074057/219499.77
 = 9.45

K1 = 3 (e - 0.5 • Bplen)
 = 3 (195.44 - 0.5•720.0)
 = -493.68 mm

K2 = 6 • n1 • At / Bpwid • (f + e)
 = 6 • 9.45 • 4.23/150.0 • (310.0 + 195.44)
 = 80903.03 mm²

K3 = -K2 • (0.5 • Bplen + f)
 = -80903.03 • (0.5 • 720.0 + 310.0)
 = -54205036.26 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -493.68 \cdot Y^2 + 80903.03 \cdot Y + -0.5E+08 = 0$
 Y = 532.79 mm

Num = (Bplen / 2 - Y / 3 - e)
 = (720.0/2 - 532.79/3 - 195.44)
 = -13.04

Denom = (Bplen / 2 - Y / 3 + f)
 = (720.0/2 - 532.79/3 + 310.0)
 = 492.40

Total Bolt Tension Force [Tforce]:
 = - QO • Num / Denom
 = - 2614.02 • -13.04/492.4
 = 69.20 kgf

Bolt Area Required due to Transverse Load [Bltareart]:
 = Tforce / (Stba • Nbt)
 = 69.2/(1399.95 • 1.0)
 = 0.0494 cm²

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.3569, 0.0494]
 = 0.3569 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = 2(QO + Tforce) / (Y • Bpwid)
 = 2(2614.02 + 69.2)/(532.79 • 150.0)
 = 6.71 kgf/cm²

Distance from Baseplate Edge to the Web [ADIST]:
 = (Bplen - WebIngh) / 2
 = (720.0 - 669.2)/2
 = 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:



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$$\begin{aligned}
 &= T_{force} \cdot A_{dist} \\
 &= 69.2 \cdot 2.54 \\
 &= 1.76 \text{ kgf-m.}
 \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned}
 &= f_c \cdot (Y - A_{dist}) / Y \\
 &= 6.71 \cdot (532.79 - 25.4) / 532.79 \\
 &= 6.39 \text{ kgf/cm}^2
 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned}
 &= (A_{dist}^2 \cdot B_{pwid} / 6) \cdot (f_1 + 2 \cdot f_c) \\
 &= (25.4^2 \cdot 150.0 / 6) \cdot (6.39 + 2 \cdot 6.71) \\
 &= 3.20 \text{ kgf-m.}
 \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned}
 &= \sqrt{(6 \cdot \max(M_t, M_c) / (B_{pwid} \cdot S_{ba}))} \\
 &= \sqrt{(6 \cdot \max(1.76, 3.2) / (150.0 \cdot 2676.61))} \\
 &= 2.1860 \text{ mm}
 \end{aligned}$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.383, Temp. Reduction = 55 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

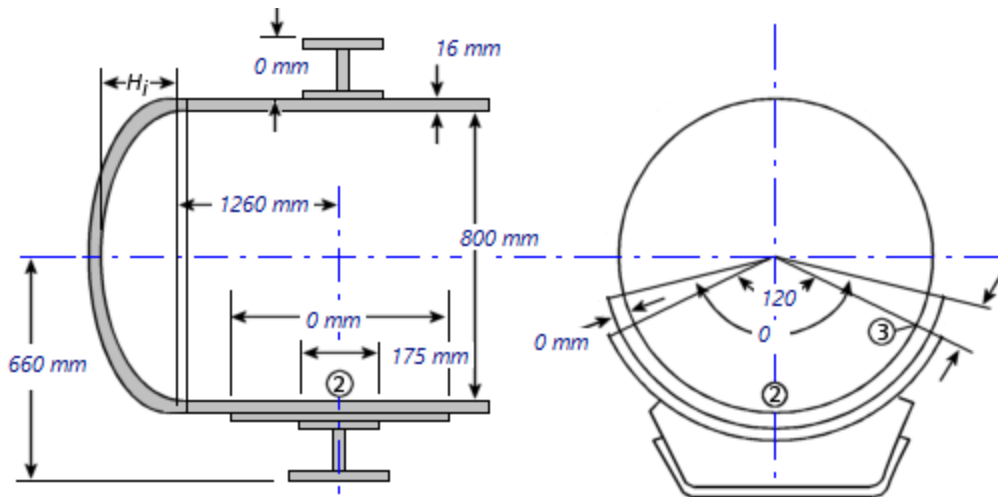
Horizontal Vessel Stress Calculations : Test Case

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{409.6 \cdot 12.8}, 2 \cdot 1260.0)$
 $= 287.9561 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	409.60	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	643.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	175.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		7974.44	kgf
Pressure used in Analysis	P	12.635	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	159.42	2386.01
Long. Stress at Bottom of Midspan	244.89	2386.01
Long. Stress at Top of Saddles	461.28	2386.01
Long. Stress at Bottom of Saddles	58.53	2386.01



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Tangential Shear in Shell	108.32	1413.05
Circ. Stress at Horn of Saddle	439.95	3579.01
Circ. Compressive Stress in Shell	16.45	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(510.9/2 + 0) \cdot 660.0/720.0$
 $= 702.5\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(57.5, 0) \cdot 660.0/3780.0007$
 $= 10.0\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 7272 + \max(10, 702, 0, 0)$
 $= 7974.4\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1984.302(0.0 + \pi/4(0.832 \cdot 1.2)^2)$
 $= 1708.845\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8076.51	kgf
Transverse Shear Load	255.44	kgf
Longitudinal Shear Load	57.50	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 720.000\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -7974 \cdot 126.0 [1 - (1 - 126.0/643.4 + (40.96^2 - 0.0^2) / (2 \cdot 126.0 \cdot 643.4)) / (1 + (4 \cdot 0.0) / (3 \cdot 643.4))]$
 $= -1863.7\ kgf \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 7974 \cdot 643 / 4 ((1 + 2(41^2 - 0^2) / (643^2)) / (1 + (4 \cdot 0) / (3 \cdot 643)) - 4 \cdot 126 / 643)$
 $= 2883.1\ kgf \cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 12.635 \cdot 409.6 / (2 \cdot 12.8) - 2883.1 / (\pi \cdot 409.6^2 \cdot 12.8)$
 $= 159.42\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$



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$$= 12.635 \cdot 409.6 / (2 \cdot 12.8) + 2883.1 / (\pi \cdot 409.6^2 \cdot 12.8)$$

$$= 244.89 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 12.635 \cdot 409.6 / (2 \cdot 12.8) - 1863.7 / (0.1066 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 461.28 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 12.635 \cdot 409.6 / (2 \cdot 12.8) + 1863.7 / (0.1923 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 58.53 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 7974(643.4 - 2 \cdot 126.0) / (643.4 + (4 \cdot 0.0 / 3))$$

$$= 4851.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4851.1 / (409.6 \cdot 12.8)$$

$$= 108.32 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(409.6 \cdot 12.8)}$$

$$= 56.478 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 7974 \cdot 0.1 / (12.8(175.0 + 56.48 + 56.48))$$

$$= -16.45 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -7974 / (4 \cdot 12.8(175.0 + 56.478 + 56.478)) - 3 \cdot 0.0529 \cdot 7974 / (2 \cdot 12.8^2)$$

$$= -439.95 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{(409.6 \cdot 12.8)}, 2 \cdot 126.0)$$

$$= 287.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(403.2 \cdot 12.8)} + 225.0)^3) / 12$$

$$= 5.891 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$



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$$= 5.891 + 43.145(88.143 - 6.4)^2$$

$$= 2888.831 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	337.1	12.8	6.4	43.1	27612.8	0.289E+04
Wearplate	225.0	12.0	18.8	27.0	50760.0	0.130E+04
Web	16.0	216.0	132.8	34.6	458956.9	0.203E+04
BasePlate	150.0	16.0	248.8	24.0	597120.0	0.620E+04
Totals	...	...	...	128.7	1134449.8	0.124E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1134450/128.705$$

$$= 88.143 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7974.441$$

$$= 1622.9718 \text{ kgf}$$

Tension Stress, $St = (Fh/As) = 18.9688 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.556 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 529.9922 \text{ kgf-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 37.6042 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(7974 + 102)150.0 / (2 \cdot 720.0 \cdot 1767.415))}$$

$$= 11.950 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 741.3177 / (3 - 1)$$

$$= 370.659 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 370.6589 \cdot 150.0/2$$



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$$= 277.994 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 8076.511 / (720.0 \cdot 150.0)$$

$$= 7.478 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 278.0 \cdot 7.48$$

$$= 2078.910 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 29.653$$

$$= 55.093 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2078.9 / 55.0927$$

$$= 37.735 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.110E+04
Web	185.3	16.0	8.0	29.7	23722.2	490.
Totals	...	...	...	55.1	266674.2	0.159E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 266674.156 / 55.093$$

$$= 48.405 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(48.405, 175.0 - 48.405)$$

$$= 126.595 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1590.6 / 55.093)}$$

$$= 53.732 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 660.0 - \cos(120.0 / 2) (400.0 + 16.0 + 12.0) - 16.0$$

$$= 430.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 720.0 - 50.8$$



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= 669.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (741.318 - 669.2)/2
 = 36.059 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(430.0^2 + 36.059^2)}$
 = 431.509 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2651.1)}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 431.509/53.732
 = 8.031

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 57.5 / (2 • 720.0) • 370.659 • 431.51/2
 = 3.194 kgf-m.

Compressive Allowable, KL/r < Cc (8.0307 < 123.2106) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (8.03)² / (2 • 123.21²)) 2651 /
 (5/3 + 3 • (8.03) / (8 • 123.21) - (8.03³) / (8 • 123.21³))
 = 1564 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 37.73/1564.39 + (3.19 • 126.595/15906263) / 1767.42
 = 0.026

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.149E+04
Web	370.7	16.0	8.0	59.3	47444.3	422.
Totals	...	...	...	84.7	290396.3	0.191E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 290396.312/84.745
 = 34.267 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))
 = max(34.267, abs(175.0 - 34.267))
 = 140.733 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 660.0 - cos(428.0/2) (16.0 + 0.0 + 0.0) - 0.0
 = 216.000 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
 = $\sqrt{(1911.7 / 84.745)}$



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= 47.495 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 216.0/47.495
 = 4.548

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 57.504/(2 • 720.0)
 = 0.040 kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.04 • 370.659 • 216.0
 = 3.197 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 3.197/135835.672
 = 2.354 kgf/cm²

Compressive Allowable, KL/r < Cc (4.5479 < 123.2106) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(4.55)²/(2 • 123.21²))2651/
 (5/3+3•(4.55)/(8• 123.21)-(4.55³)/(8•123.21³))
 = 1577 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 54.22/1576.5 + (3.2 • 140.733/1911.656)/1767.42
 = 0.036

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	7374.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	10.0 kgf
Maximum Transverse Force	Ft	255.4 kgf
Maximum Longitudinal Force	Fl	57.5 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)
 = 57.5/(4.235 • 2)
 = 6.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 255.4/(4.235 • 2)
 = 30.2 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)



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Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 57.5 / (1052.29 \cdot 2.0)$
 $= 0.0273 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 660.0 \cdot 255.44 + 0.0$
 $= 168.59 \text{ kgf-m.}$

Eccentricity (e):
 $= \text{Rmom} / \text{QO}$
 $= 168.59 / 7374.05$
 $= 22.86 \text{ mm} < \text{Bplen} / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 0.0273, 0.0]$
 $= 0.0273 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{\text{rm} \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ \text{Rm} \cdot t }, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{ 409.6 \cdot 12.8 }, 2 \cdot 1260.0)$
 $= 287.9561 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	409.60	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	643.40	cm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	175.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	20.67	cm
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		4090.38	kgf
Pressure used in Analysis	P	12.635	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual	Allowable
	kgf/cm ²	kgf/cm ²



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Long. Stress at Top of Midspan	184.47	2386.01
Long. Stress at Bottom of Midspan	219.84	2386.01
Long. Stress at Top of Saddles	360.85	2386.01
Long. Stress at Bottom of Saddles	114.20	2386.01
Tangential Shear in Shell	53.28	1413.05
Circ. Stress at Horn of Saddle	225.67	3579.01
Circ. Compressive Stress in Shell	8.44	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(510.9/2 + 0) \cdot 660.0/720.0$
 $= 702.5 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(57.5, 0) \cdot 660.0/3780.0007$
 $= 10.0 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 3388 + \max(10, 702, 0, 0)$
 $= 4090.4 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.302(0.0 + \pi/4(0.832 \cdot 1.2)^2)$
 $= 1708.845 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4192.45	kgf
Transverse Shear Load	255.44	kgf
Longitudinal Shear Load	57.50	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 720.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -4090 \cdot 126.0 [1 - (1 - 126.0/643.4 + (40.96^2 - 20.67^2)/(2 \cdot 126.0 \cdot 643.4))/(1 + (4 \cdot 20.67)/(3 \cdot 643.4))]$
 $= -1141.4 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 4090 \cdot 643/4((1 + 2(41^2 - 21^2)/(643^2))/(1 + (4 \cdot 21)/(3 \cdot 643)) - 4 \cdot 126/643)$
 $= 1193.4 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 12.635 \cdot 409.6/(2 \cdot 12.8) - 1193.4/(\pi \cdot 409.6^2 \cdot 12.8)$



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$$= 184.47 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) $[\sigma_2]$:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 12.635 \cdot 409.6 / (2 \cdot 12.8) + 1193.4 / (\pi \cdot 409.6^2 \cdot 12.8)$$

$$= 219.84 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) $[\sigma_{*3}]$:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 12.635 \cdot 409.6 / (2 \cdot 12.8) - 1141.4 / (0.1066 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 360.85 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) $[\sigma_{*4}]$:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 12.635 \cdot 409.6 / (2 \cdot 12.8) + 1141.4 / (0.1923 \cdot \pi \cdot 409.6^2 \cdot 12.8)$$

$$= 114.20 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) $[T]$:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 4090(643.4 - 2 \cdot 126.0) / (643.4 + (4 \cdot 20.67 / 3))$$

$$= 2386.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) $[\tau_2]$:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2386.09 / (409.6 \cdot 12.8)$$

$$= 53.28 \text{ kgf/cm}^2$$

Decay Length (4.15.20) $[x_1, x_2]$:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{409.6 \cdot 12.8}$$

$$= 56.478 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) $[\sigma_6]$:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 4090 \cdot 0.1 / (12.8(175.0 + 56.48 + 56.48))$$

$$= -8.44 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) $[\sigma_7]$:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -4090 / (4 \cdot 12.8(175.0 + 56.478 + 56.478)) -$$

$$3 \cdot 0.0529 \cdot 4090 / (2 \cdot 12.8^2)$$

$$= -225.67 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) $[B_1]$:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{409.6 \cdot 12.8}, 2 \cdot 126.0)$$

$$= 287.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell $[ShellInertia]$:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$



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$$= ((1.56 \cdot \sqrt{(403.2 \cdot 12.8)} + 225.0) / 12.8^3) / 12$$

$$= 5.891 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 5.891 + 43.145 (88.143 - 6.4)^2$$

$$= 2888.831 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	337.1	12.8	6.4	43.1	27612.8	0.289E+04
Wearplate	225.0	12.0	18.8	27.0	50760.0	0.130E+04
Web	16.0	216.0	132.8	34.6	458956.9	0.203E+04
BasePlate	150.0	16.0	248.8	24.0	597120.0	0.620E+04
Totals	...	...	...	128.7	1134449.8	0.124E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1134450 / 128.705$$

$$= 88.143 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4090.377$$

$$= 832.4803 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 9.7298 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 660.0 - 403.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.556 \text{ mm}$$

Bending Moment, M = Fh * d = 271.8520 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 19.2885 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4090 + 102) \cdot 150.0) / (2 \cdot 720.0 \cdot 1767.415)}$$

$$= 8.610 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (400.0 + 16.0 + 12.0)$$

$$= 741.318 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 741.3177 / (3 - 1)$$



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= 370.659 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 370.6589 • 150.0/2
 = 277.994 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 4192.446 / (720.0 • 150.0)
 = 3.882 kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 278.0 • 3.88
 = 1079.144 kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 25.44 + 29.653
 = 55.093 cm²

Compressive Stress [Sc]:

= P/Ar
 = 1079.1/55.0927
 = 19.588 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.110E+04
Web	185.3	16.0	8.0	29.7	23722.2	490.
Totals	...	...	...	55.1	266674.2	0.159E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 175.0 - 16.0
 = 159.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 266674.156/55.093
 = 48.405 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(48.405, 175.0 - 48.405)
 = 126.595 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1590.6/55.093}$
 = 53.732 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 660.0 - cos(120.0/2)(400.0 + 16.0 + 12.0) - 16.0
 = 430.000 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120.0/2)(400.0 + 16.0 + 12.0)



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= 741.318 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 720.0 - 50.8
 = 669.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (741.318 - 669.2)/2
 = 36.059 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(430.0^2 + 36.059^2)}$
 = 431.509 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2651.1)}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 431.509/53.732
 = 8.031

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 57.5 / (2 • 720.0) • 370.659 • 431.51/2
 = 3.194 kgf-m.

Compressive Allowable, KL/r < Cc (8.0307 < 123.2106) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (8.03)² / (2 • 123.21²)) 2651 /
 (5/3 + 3 • (8.03) / (8 • 123.21) - (8.03³) / (8 • 123.21³))
 = 1564 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 19.59/1564.39 + (3.19 • 126.595/15906263) / 1767.42
 = 0.014

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.149E+04
Web	370.7	16.0	8.0	59.3	47444.3	422.
Totals	...	...	...	84.7	290396.3	0.191E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 290396.312/84.745
 = 34.267 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))
 = max(34.267, abs(175.0 - 34.267))
 = 140.733 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 660.0 - cos(428.0/2) (16.0 + 0.0 + 0.0) - 0.0



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= 216.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1911.7/84.745}$
 = 47.495 mm

Slenderness ratio [KL/r]:
 = KL/r
 $= 1 \cdot 216.0/47.495$
 = 4.548

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 57.504 / (2 \cdot 720.0)$
 = 0.040 kgf/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force $\cdot e \cdot L$
 $= 0.04 \cdot 370.659 \cdot 216.0$
 = 3.197 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 $= 3.197/135835.672$
 = 2.354 kgf/cm²

Compressive Allowable, $KL/r < C_c$ (4.5479 < 123.2106) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (4.55)^2 / (2 \cdot 123.21^2)) 2651 /$
 $(5/3 + 3 \cdot (4.55) / (8 \cdot 123.21) - (4.55^3) / (8 \cdot 123.21^3))$
 = 1577 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 28.15/1576.5 + (3.2 \cdot 140.733/1911.656) / 1767.42$
 = 0.019

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	3490.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	10.0 kgf
Maximum Transverse Force	Ft	255.4 kgf
Maximum Longitudinal Force	F1	57.5 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 57.5 / (4.235 \cdot 2)$
 = 6.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 255.4 / (4.235 \cdot 2)$
 = 30.2 kgf/cm². Must be less than 1052.3 kgf/cm².



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Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 57.5/(1052.29 • 2.0)
= 0.0273 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 660.0 • 255.44 + 0.0
= 168.59 kgf-m.

Eccentricity (e):
= Rmom / QO
= 168.59/3489.98
= 48.31 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0273, 0.0]
= 0.0273 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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 Nozzle Calcs.: N2- Channel out Nozl: 9 9:54am Aug 12,2026

Input, Nozzle Desc: N2- Channel outlet

From: 20

Pressure for Reinforcement Calculations	P	5.625	kgf/cm ²
Temperature for Internal Pressure	Temp	71	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		35.00	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

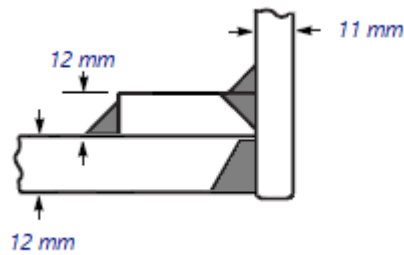


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Nozzle Calcs.: N2- Channel out Nozl: 9 9:54am Aug 12,2026



Reinforcement CALCULATION, Description: N2- Channel outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.62 \cdot 403.2 / (1407 \cdot 1.0 - 0.6 \cdot 5.62)$
 $= 1.6156 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5.62 \cdot 84.1375 / (1203 \cdot 1.0 + 0.4 \cdot 5.62)$
 $= 0.3926 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.4588	mm
Parallel to Vessel Wall, opening length	d	152.7294	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

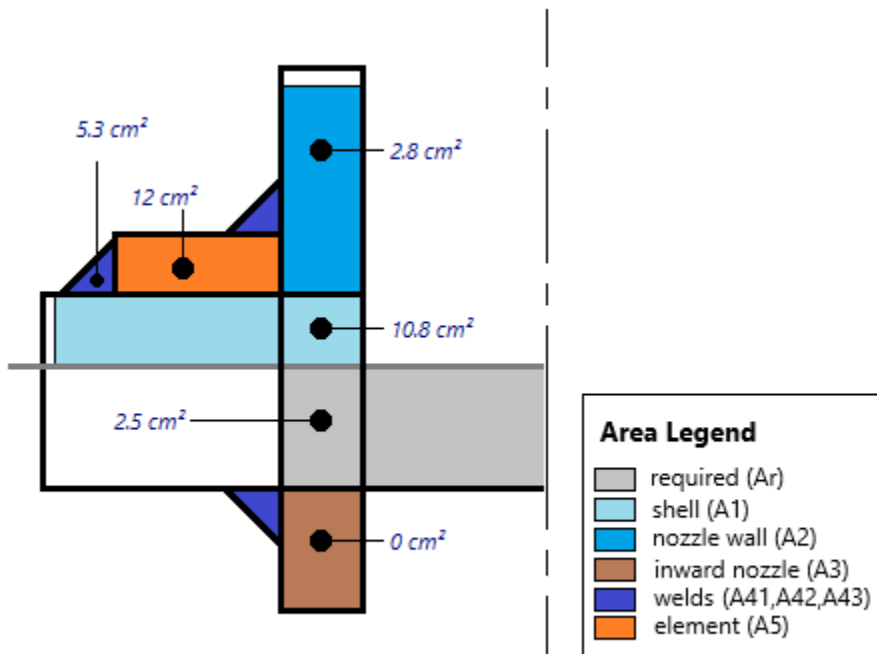


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2.504	NA	NA
Area in Shell	A1	10.811	NA	NA
Area in Nozzle Wall	A2	2.777	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.275	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	30.863	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152.7294 \cdot 1.6156 \cdot 1.0 + 2 \cdot 7.7728 \cdot 1.6156 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 2.504 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152.729(1.0 \cdot 8.8 - 1.0 \cdot 1.616) - 2 \cdot 7.773$$

$$(1.0 \cdot 8.8 - 1.0 \cdot 1.6156) \cdot (1 - 0.855)$$

$$= 10.811 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.0) \cdot (7.77 - 0.39) \cdot 0.8551$$



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= 2.777 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
 = (5.0) • 0.86 + (0.0) • 0.86 + 10.0² • 1.0
 = 5.275 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (268.275 - 168.275) • 12.0 • 1.0
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.5926 mm
 Wall Thickness per UG16(b), tr16b = 5.7000 mm
 Wall Thickness, shell/head, internal pressure trb1 = 4.8156 mm
 Wall Thickness tb1 = max(trb1, tr16b) = 5.7000 mm
 Wall Thickness tb2 = max(trb2, tr16b) = 5.7000 mm
 Wall Thickness per table UG-45 tb3 = 9.4200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[9.42, max(5.7, 5.7)]
 = 5.7000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.5926, 5.7)
 = 5.7000 mm

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	321.0,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2687.0 kgf/cm ²	Passed
Occasional	:	26.3,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	207.5,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.393, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.061, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.393, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.061, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.393, c = 3.2 mm, E* = 1.0



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 Nozzle Calcs.: N2- Channel out Nozl: 9 9:54am Aug 12,2026

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.061$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 12.0$, $tr = 1.616$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.184$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 9.601$, $tr = 0.393$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.061$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5.62 / 19.99 = 0.281$

Weld Size Calculations, Description: N2- Channel out

Intermediate Calc. for nozzle/shell Welds T_{min} 7.7728 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 8.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4410 = 0.7 \cdot t_{min}$	$21.2100 = 0.7 \cdot W_o$ mm
Pad Weld	$4.4000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.5039 - 10.8108 + 2 \cdot 7.7728 \cdot 0.8551 \cdot (1.0 \cdot 8.8 - 1.6156)) \cdot 1407)$
 $= \max(0, -10345.03) \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2.7767 + 12.0 + 5.2754 - 0.0 \cdot 0.86) \cdot 1407$
 $= 28215.86 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.7767 + 0.0 + 6.7407 + (1.1698)) \cdot 1407$
 $= 15038.16 \text{ kgf}$

Weld Load [W3]:



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$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (2.7767 + 0.0 + 5.2754 + 12.0 + (1.1698)) \cdot 1407$$

$$= 29861.85 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 30.0 \cdot 0.49 \cdot 1203$$

$$= 46751. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 268.275 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 29056. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 80.2511) \cdot (10.9728 - 3.2) \cdot 0.7 \cdot 1203$$

$$= 16505. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 1407$$

$$= 33028. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (12.0 - 3.2) \cdot 0.74 \cdot 1407$$

$$= 24221. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (29056 + 16505) = 45561 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (46751 + 33028 + 24221 + 0) = 104001 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (29056 + 24221 + 0) = 53277 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 45560	kgf, must exceed W = 0 kgf	or W1 = 28215 kgf
Path 2-2 = 104000	kgf, must exceed W = 0 kgf	or W2 = 15038 kgf
Path 3-3 = 53276	kgf, must exceed W = 0 kgf	or W3 = 29861 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 8.9490 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 270.9490 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 112.59 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 152.73 mm

Input Echo, WRC297 Item 1, Description: N2- Channel ou:



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Nozzle Calcs.: N2- Channel out Nozl: 9 9:54am Aug 12,2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	268.275	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5.62	kgf/cm ²
Radial Load	P	-407.91	kgf
Circumferential Shear	Vc	611.86	kgf
Longitudinal Shear	Vl	611.86	kgf
Circumferential Moment	Mc	275.23	kgf-m.
Longitudinal Moment	Ml	366.97	kgf-m.
Torsional Moment	Mt	458.71	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{413.6(12.0 - 3.2)}$$

$$= 367.363 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	827.200	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	20.800	mm
Nozzle Thickness used	(t) =	7.773	mm
T / t	=	2.676	
d / t	=	21.649	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.283$$

Nr/P	=	0.087
Mr/P	=	0.096



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MØ/P = 0.082
 NØ/P = 0.189
 MrD/Mc = 0.183
 NrDL/Mc = 0.134
 MØd/Mc = 0.175
 NØDL/Mc = 0.133
 MrD/Ml = 0.138
 NrDL/Ml = 0.104
 MØD/Ml = 0.130
 NØDL/Ml = 0.267

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	46	-46	46	-46
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-134	-134	134	134
Outplane Bending (ML)	-393	393	393	-393
Normal Pressure Stress	106	111	106	111
Outplane Stress Summary	-357	342	698	-175

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-52	-52	52	52
Inplane Bending (ML)	-418	418	418	-418
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	-354	372	586	-358

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	49	49	49	49
Shear Stress Summary	60	60	38	38

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	416	419	710	366

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	17	17	17	17



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 Nozzle Calcs.: N2- Channel out Nozl: 9 9:54am Aug 12,2026

Outplane Bending (P)	46	-46	46	-46
Outplane Membrane (Mc)	-50	-50	50	50
Outplane Bending (Mc)	-396	396	396	-396
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53
Outplane Stress Summary	-329	371	564	-321

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-50	-50	50	50
Inplane Bending (Mc)	-415	415	415	-415
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	106	111	106	111
Inplane Stress Summary	-296	430	634	-298

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	49	49	49	49
Shear Stress Summary	38	38	60	60

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	355	449	669	372

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-134	-134	134	134
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	63	69	63	69
Hoop Stress Summary	-53	-47	215	222

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	213	-213	213	-213
Axial Membrane (Mc)	0	0	0	0



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 Nozzle Calcs.: N2- Channel out Nozl: 9 9:54am Aug 12,2026

Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-244	-244	244	244
Axial Bending (ML)	-1862	1862	1862	-1862
Axial Pressure Stress	31	31	31	31

Axial Stress Summary	-1851	1447	2361	-1789
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	132	132	132	132
Shear Stress Summary	162	162	102	102

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1866	1529	2366	2022

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-50	-50	50	50
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	63	69	63	69
Hoop Stress Summary	30	37	131	137

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	213	-213	213	-213
Axial Membrane (Mc)	-183	-183	183	183
Axial Bending (Mc)	-1883	1883	1883	-1883
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	31	31	31	31
Axial Stress Summary	-1811	1529	2321	-1871

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	132	132	132	132
Shear Stress Summary	102	102	162	162

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl



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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1853	1536	2333	2035

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(407.6(12.0 - 3.2))}$$

$$= 365.914 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	815.200 mm
Nozzle Outside Diameter	(d) =	268.275 mm
Vessel Thickness used	(T) =	8.800 mm
Nozzle Thickness used	(t) =	57.773 mm
T / t	=	0.152
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.167$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.059
Mr/P	=	0.102
M0/P	=	0.030
N0/P	=	0.051
MrD/Mc	=	0.265
NrDL/Mc	=	0.100
M0d/Mc	=	0.079
N0DL/Mc	=	0.074
MrD/Ml	=	0.078
NrDL/Ml	=	0.069
M0D/Ml	=	0.023
N0DL/Ml	=	0.061

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	95	-95	95	-95
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-108	-108	108	108
Outplane Bending (ML)	-241	241	241	-241
Normal Pressure Stress	254	260	254	260
Outplane Stress Summary	26	324	726	59

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	322	-322	322	-322
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-122	-122	122	122
Inplane Bending (ML)	-825	825	825	-825
Inplane Pressure Stress	127	127	127	127



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Inplane Stress Summary	-467	538	1428	-866
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	16	16	-16	-16
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	46	46	46	46

Shear Stress Summary	62	62	29	29
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	509	555	1429	928
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	95	-95	95	-95
Outplane Membrane (Mc)	-98	-98	98	98
Outplane Bending (Mc)	-631	631	631	-631
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	127	127	127	127

Outplane Stress Summary	-480	591	978	-473
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	322	-322	322	-322
Inplane Membrane (Mc)	-131	-131	131	131
Inplane Bending (Mc)	-2103	2103	2103	-2103
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	254	260	254	260

Inplane Stress Summary	-1627	1940	2843	-2002
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-16	-16	16	16
Torsional Shear (Mt)	46	46	46	46

Shear Stress Summary	29	29	62	62
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1628	1941	2845	2005

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	106	111	106	111	106	111	106	111
Circ. Pl (SUS)	-117	-117	151	151	-42	-42	58	58
Circ. Q (SUS)	-347	347	439	-439	-361	361	469	-469
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-44	-44	60	60	-33	-33	67	67
Long. Q (SUS)	-364	364	472	-472	-350	350	442	-442
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	49	49	49	49	49	49	49	49
Pm (SUS)	106.0	111.0	106.0	111.0	106.0	111.0	106.0	111.0
Pm+Pl (SUS)	29.7	26.6	257.8	262.8	66.6	71.4	166.6	171.4
Pm+Pl+Q (Total)	416.5	419.1	707.8	366.6	354.9	448.4	667.2	372.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	111.00	1407.13	Passed
Pm+Pl (SUS)	262.81	2110.70	Passed
Pm+Pl+Q (TOTAL)	707.76	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	254	260	254	260	254	260	254	260
Circ. Pl (SUS)	-82	-82	134	134	-101	-101	161	161
Circ. Q (SUS)	-146	146	336	-336	-1781	1781	2425	-2425
Long. Pm (SUS)	127	127	127	127	127	127	127	127
Long. Pl (SUS)	-92	-92	152	152	-72	-72	124	124
Long. Q (SUS)	-503	503	1147	-1147	-536	536	726	-726
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-16	-16	16	16
Shear Q (SUS)	46	46	46	46	46	46	46	46
Pm (SUS)	254.0	260.0	254.0	260.0	254.0	260.0	254.0	260.0
Pm+Pl (SUS)	173.8	179.8	390.3	396.2	155.5	161.4	416.5	422.5



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Pm+Pl+Q (Total)	509	554	1427	927	1628	1940	2842	2006
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Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	260.00	1407.13	Passed
Pm+Pl (SUS)	422.49	2110.70	Passed
Pm+Pl+Q (TOTAL)	2842.06	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	69	63	69	63	69	63	69
Circ. Pl (SUS)	-117	-117	151	151	-33	-33	67	67
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-234	-234	254	254	-173	-173	193	193
Long. Q (SUS)	-1649	1649	2075	-2075	-1670	1670	2096	-2096
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	132	132	132	132	132	132	132	132
Pm (SUS)	63.0	69.0	63.0	69.0	63.0	69.0	63.0	69.0
Pm+Pl (SUS)	208.4	208.2	295.3	296.1	181.5	187.2	232.2	232.7
Pm+Pl+Q (Total)	1866	1528	2364	2020	1853	1535	2331	2033

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	69.00	1203.20	Passed
Pm+Pl (SUS)	296.06	1804.80	Passed
Pm+Pl+Q (TOTAL)	2364.93	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1- Channel inlet

From: 20

Pressure for Reinforcement Calculations	P	5.705	kgf/cm ²
Temperature for Internal Pressure	Temp	71	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		35.00	cm
User Entered Minimum Design Metal Temperature		11.40	°C

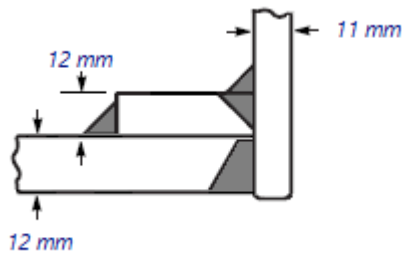
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N1- Channel inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.7 \cdot 403.2 / (1407 \cdot 1.0 - 0.6 \cdot 5.7)$
 $= 1.6387 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5.7 \cdot 84.1375 / (1203 \cdot 1.0 + 0.4 \cdot 5.7)$
 $= 0.3982 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.4588	mm
Parallel to Vessel Wall, opening length	d	152.7294	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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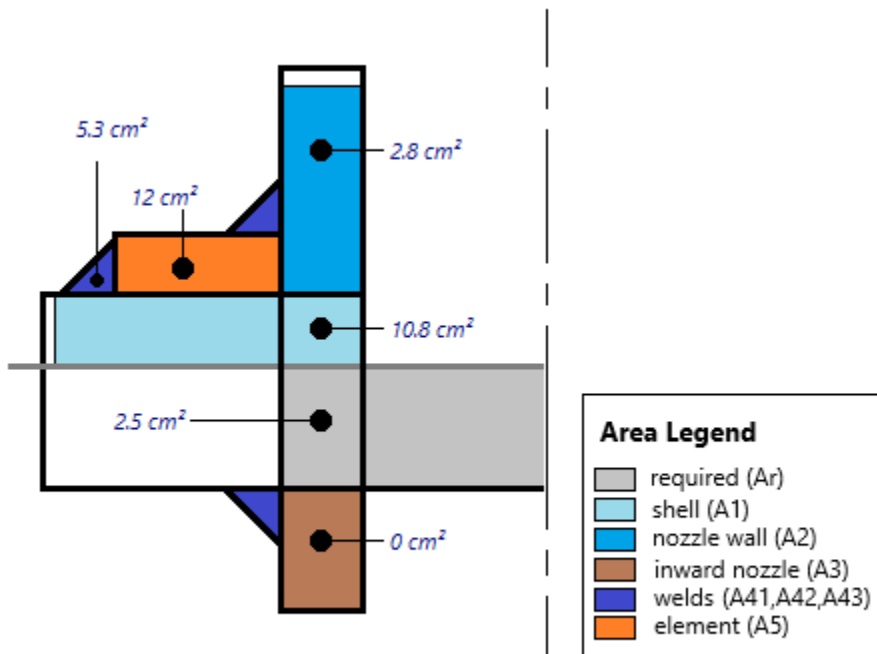


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2.540	NA	NA
Area in Shell	A1	10.776	NA	NA
Area in Nozzle Wall	A2	2.775	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.275	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	30.826	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152.7294 \cdot 1.6387 \cdot 1.0 + 2 \cdot 7.7728 \cdot 1.6387 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 2.540 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152.729(1.0 \cdot 8.8 - 1.0 \cdot 1.639) - 2 \cdot 7.773$$

$$(1.0 \cdot 8.8 - 1.0 \cdot 1.6387) \cdot (1 - 0.855)$$

$$= 10.776 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.0) \cdot (7.77 - 0.4) \cdot 0.8551$$



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= 2.775 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
 = (5.0) • 0.86 + (0.0) • 0.86 + 10.0² • 1.0
 = 5.275 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (268.275 - 168.275) • 12.0 • 1.0
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.5982 mm
 Wall Thickness per UG16(b), tr16b = 5.7000 mm
 Wall Thickness, shell/head, internal pressure trb1 = 4.8387 mm
 Wall Thickness tb1 = max(trb1, tr16b) = 5.7000 mm
 Wall Thickness tb2 = max(trb2, tr16b) = 5.7000 mm
 Wall Thickness per table UG-45 tb3 = 9.4200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[9.42, max(5.7, 5.7)]
 = 5.7000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.5982, 5.7)
 = 5.7000 mm

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	321.4,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2686.7 kgf/cm ²	Passed
Occasional	:	26.7,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	207.5,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.398, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.062, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.398, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.062, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.398, c = 3.2 mm, E* = 1.0



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Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 12.0$, $tr = 1.639$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.186$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 9.601$, $tr = 0.398$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.062$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5.70 / 19.99 = 0.285$

Weld Size Calculations, Description: N1- Channel inl

Intermediate Calc. for nozzle/shell Welds T_{min} 7.7728 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 8.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4410 = 0.7 \cdot t_{min}$	$21.2100 = 0.7 \cdot W_o$ mm
Pad Weld	$4.4000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.5396 - 10.7761 + 2 \cdot 7.7728 \cdot 0.8551 \cdot (1.0 \cdot 8.8 - 1.6387)) \cdot 1407)$
 $= \max(0, -10250.37)$ kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2.7746 + 12.0 + 5.2754 - 0.0 \cdot 0.86) \cdot 1407$
 $= 28212.90$ kgf

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.7746 + 0.0 + 6.7407 + (1.1698)) \cdot 1407$
 $= 15035.21$ kgf

Weld Load [W3]:



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$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (2.7746 + 0.0 + 5.2754 + 12.0 + (1.1698)) \cdot 1407$$

$$= 29858.90 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 30.0 \cdot 0.49 \cdot 1203$$

$$= 46751. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 268.275 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 29056. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 80.2511) \cdot (10.9728 - 3.2) \cdot 0.7 \cdot 1203$$

$$= 16505. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 1407$$

$$= 33028. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (12.0 - 3.2) \cdot 0.74 \cdot 1407$$

$$= 24221. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (29056 + 16505) = 45561 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (46751 + 33028 + 24221 + 0) = 104001 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (29056 + 24221 + 0) = 53277 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 45560	kgf, must exceed W = 0 kgf	or W1 = 28212 kgf
Path 2-2 = 104000	kgf, must exceed W = 0 kgf	or W2 = 15035 kgf
Path 3-3 = 53276	kgf, must exceed W = 0 kgf	or W3 = 29858 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 8.9490 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 270.9490 mm

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.2000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	800.000 mm



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Shell Thickness		12.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.2000	mm
Nozzle Diameter		168.275	mm
Nozzle Thickness		10.9728	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad		268.275	mm
Thickness of Reinforcing Pad		12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5.70	kgf/cm ²
Radial Load	P	-407.91	kgf
Circumferential Shear	Vc	611.86	kgf
Longitudinal Shear	Vl	611.86	kgf
Circumferential Moment	Mc	275.23	kgf-m.
Longitudinal Moment	Ml	366.97	kgf-m.
Torsional Moment	Mt	458.71	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{413.6(12.0 - 3.2)}$$

$$= 367.363 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	827.200	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	20.800	mm
Nozzle Thickness used	(t) =	7.773	mm
T / t	=	2.676	
d / t	=	21.649	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.283$$

Nr/P	=	0.087
Mr/P	=	0.096
M0/P	=	0.082
N0/P	=	0.189
MrD/Mc	=	0.183
NrDL/Mc	=	0.134



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MØd/Mc = 0.175
 NØDL/Mc = 0.133
 MrD/Ml = 0.138
 NrDL/Ml = 0.104
 MØD/Ml = 0.130
 NØDL/Ml = 0.267

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	46	-46	46	-46
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-134	-134	134	134
Outplane Bending (ML)	-393	393	393	-393
Normal Pressure Stress	107	113	107	113
Outplane Stress Summary	-356	343	699	-173

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-52	-52	52	52
Inplane Bending (ML)	-418	418	418	-418
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	-354	373	587	-357

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	49	49	49	49
Shear Stress Summary	60	60	38	38

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	415	420	711	365

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	46	-46	46	-46
Outplane Membrane (Mc)	-50	-50	50	50
Outplane Bending (Mc)	-396	396	396	-396
Outplane Membrane (ML)	0	0	0	0



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Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53

Outplane Stress Summary	-329	371	564	-321
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-50	-50	50	50
Inplane Bending (Mc)	-415	415	415	-415
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	113	107	113

Inplane Stress Summary	-295	431	636	-297
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	49	49	49	49

Shear Stress Summary	38	38	60	60
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	354	450	670	370
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-134	-134	134	134
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	63	70	63	70

Hoop Stress Summary	-52	-46	216	223
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	10	10	10	10
Axial Bending (P)	213	-213	213	-213
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-244	-244	244	244
Axial Bending (ML)	-1862	1862	1862	-1862
Axial Pressure Stress	31	31	31	31



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Axial Stress Summary	-1851	1447	2361	-1789
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	132	132	132	132

Shear Stress Summary	162	162	102	102
----------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1865	1528	2366	2022
------------------------	------	------	------	------

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-50	-50	50	50
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	63	70	63	70

Hoop Stress Summary	31	38	132	138
---------------------	----	----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	10	10	10	10
Axial Bending (P)	213	-213	213	-213
Axial Membrane (Mc)	-183	-183	183	183
Axial Bending (Mc)	-1883	1883	1883	-1883
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	31	31	31	31

Axial Stress Summary	-1810	1529	2321	-1870
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	132	132	132	132

Shear Stress Summary	102	102	162	162
----------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1853	1536	2333	2035
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 Nozzle Calcs.: N1- Channel inl Nozl: 10 9:54am Aug 12,2026

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{407.6(12.0 - 3.2)}$$

$$= 365.914 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 815.200 mm
 Nozzle Outside Diameter (d) = 268.275 mm
 Vessel Thickness used (T) = 8.800 mm
 Nozzle Thickness used (t) = 57.773 mm
 T / t = 0.152
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.167$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.059
 Mr/P = 0.102
 M0/P = 0.030
 N0/P = 0.051
 MrD/Mc = 0.265
 NrDL/Mc = 0.100
 M0d/Mc = 0.079
 N0DL/Mc = 0.074
 MrD/Ml = 0.078
 NrDL/Ml = 0.069
 M0D/Ml = 0.023
 N0DL/Ml = 0.061

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	26	26	26	26
Outplane Bending (P)	95	-95	95	-95
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-108	-108	108	108
Outplane Bending (ML)	-241	241	241	-241
Normal Pressure Stress	258	264	258	264
Outplane Stress Summary	30	328	729	63

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	30	30	30	30
Inplane Bending (P)	322	-322	322	-322
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-122	-122	122	122
Inplane Bending (ML)	-825	825	825	-825
Inplane Pressure Stress	129	129	129	129
Inplane Stress Summary	-465	540	1430	-865

Vessel Stresses (kgf/cm²):



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LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			16	16	-16	-16
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			46	46	46	46
Shear Stress Summary			62	62	29	29

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			511	557	1431	930

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			26	26	26	26
Outplane Bending (P)			95	-95	95	-95
Outplane Membrane (Mc)			-98	-98	98	98
Outplane Bending (Mc)			-631	631	631	-631
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			129	129	129	129
Outplane Stress Summary			-478	593	980	-471

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			30	30	30	30
Inplane Bending (P)			322	-322	322	-322
Inplane Membrane (Mc)			-131	-131	131	131
Inplane Bending (Mc)			-2103	2103	2103	-2103
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			258	264	258	264
Inplane Stress Summary			-1624	1944	2847	-1999

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-16	-16	16	16
Torsional Shear (Mt)			46	46	46	46
Shear Stress Summary			29	29	62	62

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			1624	1945	2849	2001



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	107	113	107	113	107	113	107	113
Circ. Pl (SUS)	-117	-117	151	151	-42	-42	58	58
Circ. Q (SUS)	-347	347	439	-439	-361	361	469	-469
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-44	-44	60	60	-33	-33	67	67
Long. Q (SUS)	-364	364	472	-472	-350	350	442	-442
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	49	49	49	49	49	49	49	49
Pm (SUS)	107.0	113.0	107.0	113.0	107.0	113.0	107.0	113.0
Pm+Pl (SUS)	29.1	25.6	258.8	264.8	67.5	73.3	167.5	173.3
Pm+Pl+Q (Total)	416.0	419.8	708.7	366.5	354.6	450.0	668.0	371.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	1407.13	Passed
Pm+Pl (SUS)	264.80	2110.70	Passed
Pm+Pl+Q (TOTAL)	708.68	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	258	264	258	264	258	264	258	264
Circ. Pl (SUS)	-82	-82	134	134	-101	-101	161	161
Circ. Q (SUS)	-146	146	336	-336	-1781	1781	2425	-2425
Long. Pm (SUS)	129	129	129	129	129	129	129	129
Long. Pl (SUS)	-92	-92	152	152	-72	-72	124	124
Long. Q (SUS)	-503	503	1147	-1147	-536	536	726	-726
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	16	16	-16	-16	-16	-16	16	16
Shear Q (SUS)	46	46	46	46	46	46	46	46
Pm (SUS)	258.0	264.0	258.0	264.0	258.0	264.0	258.0	264.0
Pm+Pl (SUS)	177.8	183.7	394.3	400.1	159.5	165.4	420.5	426.5
Pm+Pl+Q (Total)	511	556	1429	929	1624	1944	2846	2002

Vessel Stress Summation Comparison (kgf/cm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	264.00	1407.13	Passed
Pm+Pl (SUS)	426.48	2110.70	Passed
Pm+Pl+Q (TOTAL)	2846.06	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	70	63	70	63	70	63	70
Circ. Pl (SUS)	-117	-117	151	151	-33	-33	67	67
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-234	-234	254	254	-173	-173	193	193
Long. Q (SUS)	-1649	1649	2075	-2075	-1670	1670	2096	-2096
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	132	132	132	132	132	132	132	132
Pm (SUS)	63.0	70.0	63.0	70.0	63.0	70.0	63.0	70.0
Pm+Pl (SUS)	208.4	208.2	295.3	296.2	181.5	188.2	232.2	232.8
Pm+Pl+Q (Total)	1866	1527	2364	2021	1853	1535	2331	2034

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	70.00	1203.20	Passed
Pm+Pl (SUS)	296.19	1804.80	Passed
Pm+Pl+Q (TOTAL)	2364.93	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 Nozzle Calcs.: Shell inlet Nozl: 11 9:54am Aug 12,2026

Input, Nozzle Desc: Shell inlet

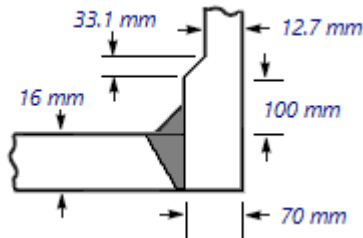
From: 50

Pressure for Reinforcement Calculations	P	9.688	kgf/cm ²
Temperature for Internal Pressure	Temp	127	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		153.65	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.0822	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.69 \cdot 403.2 / (1407 \cdot 1.0 - 0.6 \cdot 9.69)$
 $= 2.7874 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 9.69 \cdot 203.2 / (1407 \cdot 1.0 + 0.4 \cdot 9.69)$
 $= 1.3951 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 57.3 / 0.5773$
 $= 99.2465 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	774.8000 mm
Parallel to Vessel Wall, opening length	d	387.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

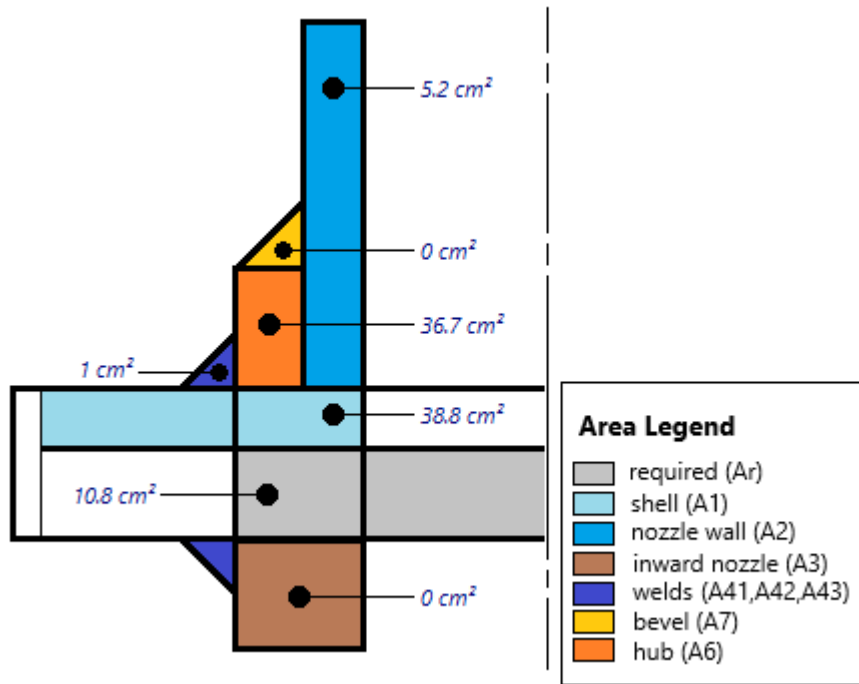


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Nozzle Calcs.: Shell inlet Nozl: 11 9:54am Aug 12,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.798	NA	NA
Area in Shell	A1	38.789	NA	NA
Area in Nozzle Wall	A2	5.187	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	36.672	NA	NA
TOTAL AREA AVAILABLE	Atot	81.648	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (387.4 \cdot 2.7874 \cdot 1.0 + 2 \cdot 66.8 \cdot 2.7874 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 10.798 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 387.4(1.0 \cdot 12.8 - 1.0 \cdot 2.787) - 2 \cdot 66.8$$

$$(1.0 \cdot 12.8 - 1.0 \cdot 2.7874) \cdot (1 - 1.0)$$

$$= 38.789 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 32.0)(9.5 - 1.4)1.0$$

$$= 5.187 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{inp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(32.0, 250.0, 100.0)) \cdot (70.0 - 12.7) \cdot 1.0$$

$$= 36.672 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5951 mm
Wall Thickness per UG16(b),	tr16b = 4.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.9874 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.9874 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.7000 mm
Wall Thickness per table UG-45	tb3 = 11.5312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.531, \max(5.9874, 4.7)]$$

$$= 5.9874 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.5951, 5.9874)$$

$$= 5.9874 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 371.1,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3146.7 kgf/cm ²	Passed
Occasional	: 96.4,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 185.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.113, tr = 1.395, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 2.787, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.218$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 9.69/19.99 = 0.485

Weld Size Calculations, Description: Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (10.7983 - 38.7889 + 2 \cdot 66.8 \cdot 1.0 \cdot (1.0 \cdot 12.8 - 2.7874)) \cdot 1407)$
 $= \max(0, -20563.46) \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (41.8591 + 1. - 0.0 \cdot 1.0) \cdot 1407$
 $= 60308.46 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (41.8591 + 0.0 + 1. + (17.1008)) \cdot 1407$
 $= 84371.55 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (41.8591 + 0.0 + 1. + 0.0 + (17.1008)) \cdot 1407$
 $= 84371.55 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 44015. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 198.45) \cdot (70.0 - 3.2) \cdot 0.7 \cdot 1407$
 $= 410214. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 406.4 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 1407$
 $= 39883. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (44015 + 410214) = 454229 \text{ kgf}$



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Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (44015 + 0 + 39883 + 0) = 83899 kgf

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (44015 + 39883 + 0) = 83899 kgf

Summary of Failure Path Calculations:

Path 1-1 = 454229 kgf, must exceed W = 0 kgf	or W1 = 60308 kgf
Path 2-2 = 83898 kgf, must exceed W = 0 kgf	or W2 = 84371 kgf
Path 3-3 = 83898 kgf, must exceed W = 0 kgf	or W3 = 84371 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 27 kgf/cm²

The Drop for this Nozzle is : 55.4572 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 321.4572 mm

Input Echo, WRC297 Item 1, Description: Shell inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	521.000	mm
Nozzle Thickness	70.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	9.69	kgf/cm ²
Radial Load	P	-1101.35	kgf
Circumferential Shear	Vc	1631.62	kgf
Longitudinal Shear	Vl	1631.62	kgf
Circumferential Moment	Mc	1957.17	kgf-m.
Longitudinal Moment	Ml	2609.55	kgf-m.
Torsional Moment	Mt	3261.93	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.626) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (7.443) is < 10



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check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 521.0 + 2 \cdot 1.65 \cdot \sqrt{(409.6(16.0 - 3.2))}$$

$$= 759.945 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 819.200 mm
 Nozzle Outside Diameter (d) = 521.000 mm
 Vessel Thickness used (T) = 12.800 mm
 Nozzle Thickness used (t) = 66.800 mm
 T / t = 0.192
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.088$$

Nr/P = 0.041
 Mr/P = 0.068
 M0/P = 0.022
 N0/P = 0.045
 MrD/Mc = 0.245
 NrDL/Mc = 0.086
 M0d/Mc = 0.073
 N0DL/Mc = 0.078
 MrD/Ml = 0.057
 NrDL/Ml = 0.039
 M0D/Ml = 0.018
 N0DL/Ml = 0.040

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	30	30	30	30
Outplane Bending (P)	87	-87	87	-87
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-121	-121	121	121
Outplane Bending (ML)	-325	325	325	-325
Normal Pressure Stress	300	310	300	310
Outplane Stress Summary	-29	456	865	49

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	27	27	27	27
Inplane Bending (P)	275	-275	275	-275
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-117	-117	117	117
Inplane Bending (ML)	-1039	1039	1039	-1039
Inplane Pressure Stress	150	150	150	150
Inplane Stress Summary	-704	824	1611	-1019

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			15	15	-15	-15
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			59	59	59	59
Shear Stress Summary			75	75	44	44

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			712	839	1613	1072

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			30	30	30	30
Outplane Bending (P)			87	-87	87	-87
Outplane Membrane (Mc)			-179	-179	179	179
Outplane Bending (Mc)			-1007	1007	1007	-1007
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			150	150	150	150
Outplane Stress Summary			-918	921	1454	-734

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			27	27	27	27
Inplane Bending (P)			275	-275	275	-275
Inplane Membrane (Mc)			-196	-196	196	196
Inplane Bending (Mc)			-3366	3366	3366	-3366
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			300	310	300	310
Inplane Stress Summary			-2959	3232	4166	-3107

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-15	-15	15	15
Torsional Shear (Mt)			59	59	59	59
Shear Stress Summary			44	44	75	75

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			2960	3233	4168	3109

Nozzle Stresses (kgf/cm²):



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LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	30	30	30	30
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-121	-121	121	121
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	40	28	40
Hoop Stress Summary	-62	-51	181	192

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-25	25	25	-25
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-29	6	74	10

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	14	14	8	8

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	68	64	181	193

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	30	30	30	30
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-179	-179	179	179
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	40	28	40
Hoop Stress Summary	-120	-108	238	250

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-101	101	101	-101
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-99	90	144	-73

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	14	14

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	123	199	240	324

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	300	310	300	310	300	310	300	310
Circ. Pl (SUS)	-91	-91	151	151	-169	-169	223	223
Circ. Q (SUS)	-238	238	412	-412	-3091	3091	3641	-3641
Long. Pm (SUS)	150	150	150	150	150	150	150	150
Long. Pl (SUS)	-90	-90	144	144	-149	-149	209	209
Long. Q (SUS)	-764	764	1314	-1314	-920	920	1094	-1094
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	15	15	-15	-15	-15	-15	15	15
Shear Q (SUS)	59	59	59	59	59	59	59	59
Pm (SUS)	300.0	310.0	300.0	310.0	300.0	310.0	300.0	310.0
Pm+Pl (SUS)	210.5	220.4	452.4	462.3	133.4	143.2	524.4	534.3
Pm+Pl+Q (Total)	712	838	1610	1072	2960	3232	4166	3110

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	310.00	1407.13	Passed
Pm+Pl (SUS)	534.28	2110.70	Passed
Pm+Pl+Q (TOTAL)	4166.02	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	40	28	40	28	40	28	40
Circ. Pl (SUS)	-91	-91	151	151	-149	-149	209	209
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-26	-26	28	28	-19	-19	21	21
Long. Q (SUS)	-18	18	32	-32	-94	94	108	-108
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	28.0	40.0	28.0	40.0	28.0	40.0	28.0	40.0
Pm+Pl (SUS)	63.1	51.1	179.0	191.0	121.0	109.0	237.0	249.0
Pm+Pl+Q (Total)	67.5	62.6	179.8	191.4	124.2	198.8	238.8	323.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	1407.13	Passed
Pm+Pl (SUS)	249.02	2110.70	Passed
Pm+Pl+Q (TOTAL)	323.05	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	9.688	kgf/cm ²
Temperature for Internal Pressure	Temp	127	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		693.65	cm
User Entered Minimum Design Metal Temperature		11.40	°C

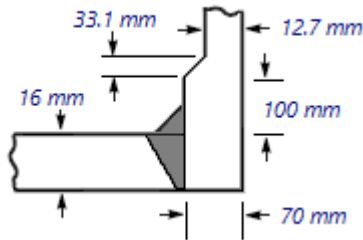
Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.0822	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.69 \cdot 403.2 / (1407 \cdot 1.0 - 0.6 \cdot 9.69)$
 $= 2.7875 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 9.69 \cdot 203.2 / (1407 \cdot 1.0 + 0.4 \cdot 9.69)$
 $= 1.3952 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 57.3 / 0.5773$
 $= 99.2465 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	774.8000 mm
Parallel to Vessel Wall, opening length	d	387.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.0000 mm

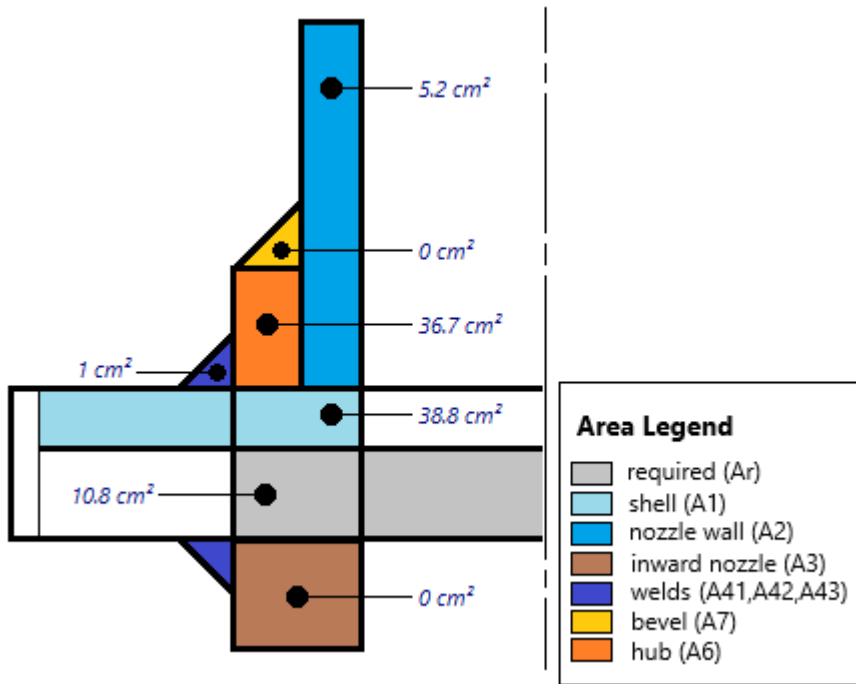
Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.799	NA	NA
Area in Shell	A1	38.788	NA	NA
Area in Nozzle Wall	A2	5.187	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	36.672	NA	NA
TOTAL AREA AVAILABLE	Atot	81.648	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (387.4 \cdot 2.7875 \cdot 1.0 + 2 \cdot 66.8 \cdot 2.7875 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 10.799 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 387.4(1.0 \cdot 12.8 - 1.0 \cdot 2.787) - 2 \cdot 66.8$$

$$(1.0 \cdot 12.8 - 1.0 \cdot 2.7875) \cdot (1 - 1.0)$$

$$= 38.788 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 32.0)(9.5 - 1.4)1.0$$

$$= 5.187 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{inp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(32.0, 250.0, 100.0)) \cdot (70.0 - 12.7) \cdot 1.0$$

$$= 36.672 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5952 mm
Wall Thickness per UG16(b),	tr16b = 4.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.9875 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.9875 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.7000 mm
Wall Thickness per table UG-45	tb3 = 11.5312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.531, \max(5.9875, 4.7)]$$

$$= 5.9875 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.5952, 5.9875)$$

$$= 5.9875 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 371.1,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3146.7 kgf/cm ²	Passed
Occasional	: 96.4,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 185.6,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.113, tr = 1.395, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.176$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 2.787, c = 3.2 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.218$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 9.69/19.99 = 0.485

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (10.7988 - 38.7885 + 2 \cdot 66.8 \cdot 1.0 \cdot (1.0 \cdot 12.8 - 2.7875)) \cdot 1407)$
 $= \max(0, -20562.41) \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (41.8591 + 1. - 0.0 \cdot 1.0) \cdot 1407$
 $= 60308.42 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (41.8591 + 0.0 + 1. + (17.1008)) \cdot 1407$
 $= 84371.51 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (41.8591 + 0.0 + 1. + 0.0 + (17.1008)) \cdot 1407$
 $= 84371.51 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 44015. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 198.45) \cdot (70.0 - 3.2) \cdot 0.7 \cdot 1407$
 $= 410214. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 406.4 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 1407$
 $= 39883. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (44015 + 410214) = 454229 \text{ kgf}$



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Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (44015 + 0 + 39883 + 0) = 83899 kgf

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (44015 + 39883 + 0) = 83899 kgf

Summary of Failure Path Calculations:

Path 1-1 = 454229 kgf, must exceed W = 0 kgf	or W1 = 60308 kgf
Path 2-2 = 83898 kgf, must exceed W = 0 kgf	or W2 = 84371 kgf
Path 3-3 = 83898 kgf, must exceed W = 0 kgf	or W3 = 84371 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 27 kgf/cm²

The Drop for this Nozzle is : 55.4572 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 321.4572 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 316.67 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 387.40 mm

Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	800.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.2000	mm
Nozzle Diameter	521.000	mm
Nozzle Thickness	70.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	9.69	kgf/cm ²
Radial Load	P	-1101.35	kgf
Circumferential Shear	Vc	1631.62	kgf
Longitudinal Shear	Vl	1631.62	kgf
Circumferential Moment	Mc	1957.17	kgf-m.
Longitudinal Moment	Ml	2609.55	kgf-m.
Torsional Moment	Mt	3261.93	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



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*Warning - The ratio, Dn/Dv (.626) is > 0.5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (7.443) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 521.0 + 2 \cdot 1.65 \cdot \sqrt{409.6(16.0 - 3.2)}$
 $= 759.945 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 819.200 mm
 Nozzle Outside Diameter (d) = 521.000 mm
 Vessel Thickness used (T) = 12.800 mm
 Nozzle Thickness used (t) = 66.800 mm
 T / t = 0.192
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.088$$

Nr/P = 0.041
 Mr/P = 0.068
 M0/P = 0.022
 N0/P = 0.045
 MrD/Mc = 0.245
 NrDL/Mc = 0.086
 M0d/Mc = 0.073
 N0DL/Mc = 0.078
 MrD/Ml = 0.057
 NrDL/Ml = 0.039
 M0D/Ml = 0.018
 N0DL/Ml = 0.040

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	30	30	30	30
Outplane Bending (P)	87	-87	87	-87
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-121	-121	121	121
Outplane Bending (ML)	-325	325	325	-325
Normal Pressure Stress	300	310	300	310
Outplane Stress Summary	-29	456	865	49

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	27	27	27	27
Inplane Bending (P)	275	-275	275	-275
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-117	-117	117	117
Inplane Bending (ML)	-1039	1039	1039	-1039
Inplane Pressure Stress	150	150	150	150



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Inplane Stress Summary -704 824 1611 -1019

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE Au Al Bu Bl
 (Shear stress normal to Top Top Bottom Bottom
 longitudinal plane) Outside Inside Outside Inside

Outplane Shear (Vc)	15	15	-15	-15
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	59	59	59	59

Shear Stress Summary 75 75 44 44

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE Au Al Bu Bl
 (Stress Intensities Top Top Bottom Bottom
 Outside Inside Outside Inside

Two * Max Shear Stress 712 839 1613 1072

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses Normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Membrane (P)	30	30	30	30
Outplane Bending (P)	87	-87	87	-87
Outplane Membrane (Mc)	-179	-179	179	179
Outplane Bending (Mc)	-1007	1007	1007	-1007
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	150	150	150	150

Outplane Stress Summary -918 921 1454 -734

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses parallel to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Inplane Membrane (P)	27	27	27	27
Inplane Bending (P)	275	-275	275	-275
Inplane Membrane (Mc)	-196	-196	196	196
Inplane Bending (Mc)	-3366	3366	3366	-3366
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	300	310	300	310

Inplane Stress Summary -2959 3232 4166 -3107

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Shear stress normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-15	-15	15	15
Torsional Shear (Mt)	59	59	59	59

Shear Stress Summary 44 44 75 75

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stress Intensities Left Left Right Right
 Outside Inside Outside Inside



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Two * Max Shear Stress	2960	3233	4168	3109
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	30	30	30	30
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-121	-121	121	121
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	40	28	40
Hoop Stress Summary	-62	-51	181	192

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-25	25	25	-25
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-29	6	74	10

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	14	14	8	8

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	68	64	181	193

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	30	30	30	30
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-179	-179	179	179
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	40	28	40
Hoop Stress Summary	-120	-108	238	250

Nozzle Stresses (kgf/cm²):



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CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-101	101	101	-101
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-99	90	144	-73

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	14	14

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	123	199	240	324

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	300	310	300	310	300	310	300	310
Circ. Pl (SUS)	-91	-91	151	151	-169	-169	223	223
Circ. Q (SUS)	-238	238	412	-412	-3091	3091	3641	-3641
Long. Pm (SUS)	150	150	150	150	150	150	150	150
Long. Pl (SUS)	-90	-90	144	144	-149	-149	209	209
Long. Q (SUS)	-764	764	1314	-1314	-920	920	1094	-1094
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	15	15	-15	-15	-15	-15	15	15
Shear Q (SUS)	59	59	59	59	59	59	59	59
Pm (SUS)	300.0	310.0	300.0	310.0	300.0	310.0	300.0	310.0
Pm+Pl (SUS)	210.5	220.4	452.4	462.3	133.4	143.2	524.4	534.3
Pm+Pl+Q (Total)	712	838	1610	1072	2960	3232	4166	3110

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	310.00	1407.13	Passed
Pm+Pl (SUS)	534.28	2110.70	Passed
Pm+Pl+Q (TOTAL)	4166.02	4221.40	Passed



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Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	40	28	40	28	40	28	40
Circ. Pl (SUS)	-91	-91	151	151	-149	-149	209	209
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-26	-26	28	28	-19	-19	21	21
Long. Q (SUS)	-18	18	32	-32	-94	94	108	-108
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	28.0	40.0	28.0	40.0	28.0	40.0	28.0	40.0
Pm+Pl (SUS)	63.1	51.1	179.0	191.0	121.0	109.0	237.0	249.0
Pm+Pl+Q (Total)	67.5	62.6	179.8	191.4	124.2	198.8	238.8	323.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	1407.13	Passed
Pm+Pl (SUS)	249.02	2110.70	Passed
Pm+Pl+Q (TOTAL)	323.05	4221.40	Passed

Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	9.69 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	9.69 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00 kgf/cm ²
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	3.2000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	800.000 mm
Shell Temperature for Internal Pressure	Ts	127.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	5.62 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	5.62 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	12.0000 mm
Channel Internal Corrosion Allowance	cac	3.2000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	800.000 mm
Channel Design Temperature	TEMPC	71.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	302
Tube Wall Thickness	et	2.5200 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6096.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5996.00 mm
Design Temperature of the Tubes		127.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1639.22 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.5200 mm
Groove Weld Leg	ag	2.5200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	379.350 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	127.00 °C
Tubesheet Material		SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	100.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.2000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.2000 mm
Tubesheet Outside Diameter	A	1020.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	866.00 mm
Step 1 Depth on the Tubeside	ht1	8.00 mm
Step 2 Diameter on the Tubeside	dt2	797.00 mm
Step 2 Depth on the Tubeside	ht2	8.00 mm
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	866.00 mm
Step 1 Depth on the Shellside	hs1	8.00 mm
Step 2 Diameter on the Shellside	ds2	797.00 mm
Step 2 Depth on the Shellside	hs2	8.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		866.00 mm

Note: Tubesheet diameter is now: 866.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	845.151 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	1020.000 mm
Flange Inside Diameter	B	800.000 mm
Flange Face Outside Diameter	Fod	869.000 mm
Flange Face Inside Diameter	Fid	800.000 mm
Gasket Outside Diameter	Go	858.000 mm
Gasket Inside Diameter	Gi	832.000 mm
Small end Hub thk.	g0	16.0000 mm
Large end Hub thk.	g1	24.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	703.07 kgf/cm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	940.000 mm
Nominal Bolt Diameter	dB	25.4000 mm
Thread Series	UNC Thread Series	
Number of Bolts	n	24
Bolt Material		SA-193 B7



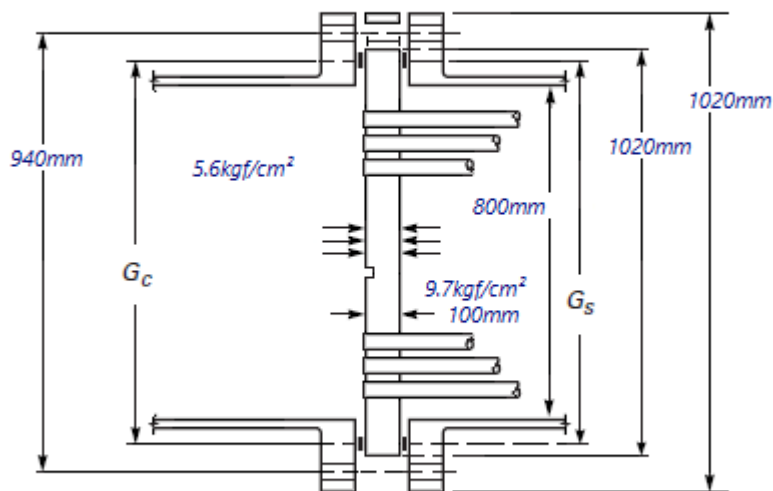
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Bolt Allowable Stress At Temperature	Sb	1753.82 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82 kgf/cm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	64261.41 kgf
Alternate Flange Seating Bolt Load, Wm2	119929.54 kgf
Alternate Flange Design Bolt Load, W	134779.06 kgf

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 68.0000 mm
Is Bolt Load Transferred to the Tubesheet	No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	127.0 °C	0.20035E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	71.1 °C	0.20378E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	127.0 °C	0.20035E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	127.0 °C	0.20035E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²



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Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5.62 \cdot (25.4/2 - 0)) / (942.17 \cdot 1.0 + 0.4 \cdot 5.62)$
 $= 0.0756 + 0.0000 = 0.0756 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 127.00 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (E_{max}):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.5200	25.40	6375.35	10.08	50.0000	0.0108275	977.27

EMA_{WP} = (4*B)/(3*(D/T)) = (4 *977.2673)/(3 *10.0794) = 129.2763 kgf/cm²

Results for Req'd Thickness for Ext. Pressure (T_{ca}):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4735	25.40	6375.35	53.64	50.0000	0.0003823	389.74

EMA_{WP} = (4*B)/(3*(D/T)) = (4 *389.7445)/(3 *53.6401) = 9.6879 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4735 mm
Allowable Internal Pressure at Corroded thickness	203.07 kgf/cm²
Required Internal Design Pressure	5.62 kgf/cm²
Allowable External Pressure at Corroded thickness	129.28 kgf/cm²
Required External Design Pressure	9.69 kgf/cm²
Required Thickness due to Shell Side pressure	0.4735 mm

Detailed Results for load Case D3 un-corr. (P_{sd,max} + P_{td,max})

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (G _{oc} - G _{ic}) / 2	13.000 mm
Basic Gasket Width,	b ₀ = N / 2	6.500 mm
Effective Gasket Width,	b = sqrt(b ₀) * 2.5	6.422 mm
Gasket Reaction Diameter,	G = G _o - 2.0*b	845.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [D_o]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 379.35 + 25.4$
 $= 784.100 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75$
 $= 0.200$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= G_s / D_o$ (Configurations d, e, f)
 $= 845.1556 / 784.1$



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$$= 1.0779$$

Ratio [pc]:

$$= G_c / D_o \text{ (Configurations d)}$$

$$= 845.1509/784.1$$

$$= 1.0779$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/100.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t(E_t/E) \cdot (StT/S) \cdot (p); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.52 (2003545/2003545) \cdot$$

$$(942/1407) \cdot (0); 25.4 - 2 \cdot 2.52)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.75 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 784.1 \cdot 31.75) / (\pi \cdot 784.1^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.2000$$

$$h/p = 3.149606 ; \mu^* = 0.200000 ; E^* = 311249 \text{ kgf/cm}^2$$

$$E^*/E = 0.155349 ; v^* = 0.458572$$

Note: As h/p (3.150) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 866.0/784.1$$

$$= 1.1045$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.46) / 311249 \cdot (2003545 \cdot \ln(1.1))$$

$$= 0.3463$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 784.1^2/16 \cdot [(1.078 - 1) \cdot (1.078^2 + 1) \cdot 9.688 -$$

$$(1.078 - 1) \cdot (1.078^2 + 1) \cdot 5.625]$$

$$= 26281.4766 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 26281.5 + 64261.4 \cdot (845.151 - 845.151) / (2 \cdot \pi \cdot 784.1)$$

$$= 26281.4766 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:



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Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((26281.48) - 784.1^2/32 \cdot 0.346 (9.69 - 5.62)) / (1 + 0.35)$$

$$= -553.6146 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= -553.61 + 784.1^2/64 \cdot (3 + 0.459) \cdot (9.69 - 5.62)$$

$$= 134426.0781 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| -553.615 | ; | 134426.078 |)$$

$$= 134426.0781 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 134426.078 / (0.2 (100.0 - 8.0)^2)$$

$$= 476.4630 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 1407.13$$

$$= 2814.27 \text{ kgf/cm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 134483.531 / (0.2 (45.9121 - 8.0)^2)$$

$$= 2806.9451 \text{ kgf/cm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 45.9121 + 0 + 0$$

$$= 45.9121 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(45.9121 ; 3.5372)$$

$$= 45.9121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 9.69 - 5.62 |$$

$$= 4.063 \text{ kgf/cm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.2 \cdot 100.0 / 784.1 \cdot \min(0.8 \cdot 1407.134 ; 0.533 \cdot 2270.603)$$

$$= 114.853 \text{ kgf/cm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu) \cdot (Do / \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2) \cdot (784.1 / 1125.707) \cdot | 9.69 - 5.62 | + 0$$

$$= 3.5372 \text{ mm}$$

Note: Analysis Completed for Tubesheet Configuration d.



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Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 2.52) 2.52$
 $= 1.811 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.52 (25.4 - 2.52) 942.168$
 $= 1706.610 \text{ kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
 $= 0.85 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 942.17$
 $= 1706.6100 \text{ kgf}$

Weld Strength Factor [fw]:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 942.168 / 942.168$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 942.17$
 $= 1111.2821 \text{ kgf}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.52 \cdot (25.4 - 2.52) 1.0 \cdot 1.0} - 0.75 \cdot 25.4 \right)$
 $= 3.111 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.5557 mm
 Minimum Required Groove Weld Leg [agr]: 1.5557 mm

Tube-Tubesheet Jt allowable, 1706.61 is >= tube strength 1706.61 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	476.5 <=	2814.3 kgf/cm ²	Ok
Tubesheet shear stress	39.8 <=	1125.7 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	45.912	100.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.556	2.520	Ok
Tube-Tubesheet Groove Weld Leg :	1.556	2.520	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Case Type	Pass/Fail
--Reqd. Thk. + CA ---- Tubesheet Stresses								



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D1uc	52.577	1.039	660	2814	55	1126	Ps+Pt D1	Ok
D2uc	66.462	1.790	1136	2814	95	1126	Ps+Pt D2	Ok
D3uc	45.912	0.751	476	2814	40	1126	Ps+Pt D3	Ok
D1c	55.755	1.039	708	2814	59	1126	Ps+Pt-c D1	Ok
D2c	69.797	1.790	1219	2814	101	1126	Ps+Pt-c D2	Ok
D3c	49.067	0.751	511	2814	43	1126	Ps+Pt-c D3	Ok

Max: 69.7968 1.790 mm 0.433 0.090 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	69.797	100.000	Ok
Tubesheet thickness flanged extension :	1.790	68.000	Ok
Tube thickness :	0.474	2.520	Ok
Tube-tubesheet fillet weld leg :	1.556	2.520	Ok
Tube-tubesheet groove weld leg :	1.556	2.520	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	22.356	1.000	22.355	1.000
Tubesheet Shear Stress	107.502	1.000	107.502	1.000
Tube Pressure Stress	203.066	1.000	129.275	1.000
Minimum MAWP	22.356		22.355	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	23.996	1.000	23.996	1.000
Tubesheet Shear Stress	114.852	1.000	114.852	1.000
Tube Pressure Stress	203.066	1.000	129.275	1.000



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Minimum MAPnc	23.996	23.996
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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4
= 100.0/4
= 25.000 mm

Thickness Ratio = 0.404, Temperature Reduction per Fig. UCS 66.1 = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(5.62/24.0, 9.69/24.0)
= 0.404

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

- ** Warning: A standard flange seems to be governing the Shellside MAWP!**
- ** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: